

A Comparative Study of Bose–Einstein Condensate and Navarro-Frenk-White Profiles using SPARC Galactic Rotation Curves Data

Tianer Tang
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Abstract

The fundamental nature of dark matter remains an open question. As dark matter modeled by the Navarro-Frenk-White (NFW) profile faces challenges such as the core-cusp problem, in this present study, we compare it against an alternative model that proposes dark matter in the form of a self-interacting Bose-Einstein Condensate (BEC), which determines the fundamental properties of dark matter by only the mass and the particle’s scattering length.

This work conducts a high-precision Bayesian model comparison using the rotation curve data of 175 galaxies from the SPARC database. Employing Markov chain Monte Carlo for posterior sampling and calculating Bayesian evidence to objectively quantify model preference, four halo models are evaluated: the standard NFW profile, a generalized NFW (gNFW) profile with a free inner slope, and both static and rotating BEC profiles.

We find that BEC-type halos, while capable of producing cored density profiles, are not systematically favored over gNFW or NFW. Cuspy NFW-like profiles remain statistically preferred for most galaxies, suggesting that the core-cusp problem can be explained phenomenologically through a continuum of inner slopes rather than requiring condensate physics.

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1 Introduction

The most widely accepted and conventional cosmological model for dark matter, the Lambda Cold Dark Matter Model(Λ CDM), describes dark matter as slow-moving, non-baryonic, collisionless, and dissipationless, constituting approximately 85% of the total matter of the universe. It interacts only through gravity and is directly undetectable due to non-interactivity with electromagnetic radiation.

Multiple lines of observational evidence point convincingly to the presence of dark matter. One of the earliest clues came from an analysis of the Coma Cluster, where an estimate of the cluster’s mass by the virial theorem yielded a significant excess compared to the visible mass in galaxies[22]. This discrepancy suggested a substantial amount of unseen matter. A similar inference was made in the 1970s when astronomers observed the rotation curves of spiral galaxies: neutral hydrogen clouds orbiting far from the galactic centers exhibited nearly constant rather than decreasing (as Newtonian dynamics would predict if only visible matter were present) rotational velocities of 100–200 km/s, i.e., as orbital velocity should decay with radius as $v(r) \propto \sqrt{GM(r)/r}$, where $M(r)$ is the enclosed mass, the observed flatness suggests a dominant dark matter halo enveloping each galaxy[18].

Further support comes from gravitational lensing, where dark matter’s gravitational field bends the light from background objects, causing distortions in their observed shapes. This effect is seen in both strong and weak lensing regimes and is particularly vivid in the Bullet Cluster. In this colliding galaxy cluster system, the hot gas clouds—which emit X-rays and represent most of the baryonic mass—lag behind the

galaxies, while gravitational lensing reveals the bulk of the mass distribution is aligned with the galaxies themselves[6]. This spatial separation between baryonic and gravitational mass distributions implies non-collisional dark matter.

On cosmological scales, observations of the cosmic microwave background (CMB) anisotropies and baryon acoustic oscillations (BAO) tightly constrain dark matter’s density and distribution, confirming its essential role in shaping large-scale structure[7]. The CMB’s acoustic peak structure, sensitive to total matter content, requires a cold, non-baryonic dark matter component, contributing about 26% of the universe’s energy density. BAO measurements reveal similar early-universe density patterns in the large-scale distribution of galaxies. The observed galaxy clustering, cosmic web structures, and formation history all point to dark matter as the gravitational framework into which baryons settled after recombination.

Despite strong indirect evidence, the nature and composition of dark matter remain key open questions. Several theoretical models have been proposed. The leading Cold Dark Matter (CDM) candidate is the Weakly Interacting Massive Particle (WIMP), a thermal relic from the early universe interacting via gravity and the weak force. Sterile neutrinos, a warm dark matter candidate, are heavier than standard neutrinos and may emit detectable X-ray decay lines. Other possibilities include primordial black holes (PBHs), fuzzy dark matter—an ultralight scalar field with wave-like behavior—and axions, proposed to solve the strong CP problem and potentially form large-scale Bose–Einstein condensates[2, 9].

One particularly intriguing alternative proposed to “go beyond” the collisionless CDM paradigm challenged by the core-cusp problem[3], the missing satellites problem, and the too-big-to-fail problem is Bose–Einstein Condensate (BEC) dark matter, which is composed of ultralight bosons that condense into a macroscopic quantum state described by a single wavefunction. This model introduces a quantum pressure term governed by the Gross–Pitaevskii equation into the dynamics, allowing dark matter to self-interact[10].

Unlike CDM, BEC dark matter naturally smooths out the central density of galactic halos, potentially resolving the long-standing core-cusp problem, which refers to the discrepancy between the “cuspy” density profiles predicted by N-body simulations under CDM (e.g., Navarro–Frenk–White or NFW profiles) and the flatter cores observed in dwarf galaxies.

In dark matter dominated galaxies, thermalization due to self-interactions creates large cores; but in more luminous galaxies, thermalization leads to denser and smaller cores, naturally explaining the flat rotation curves of the highly luminous galaxies. In particular, the wave-like nature of BEC dark matter suppresses small-scale structure formation and introduces solitonic cores in galaxy centers, which may better match observed galactic dynamics[7].

The main goal of this paper is to investigate

whether BEC dark matter provides a better phenomenological model for galactic halo structure than traditional CDM, by comparing how well analytical representations of rotational velocity of massive particles and density distributions of the halos in the framework of BEC and NFW profiles fit observational data from dark matter-dominated galaxies. Recent advances in machine learning can extract non-linear patterns from noisy astronomical data, helping to distinguish subtle differences in halo profiles. By bayesian fitting both NFW and BEC halo models to rotation curves via Bayesian methods and using machine learning for model comparison and uncertainty quantification, we assess whether BEC more accurately describes dark matter’s galactic distribution, contributing to a deeper understanding of the universe’s composition and structure formation.

After performing quality cuts on the SPARC sample to remove galaxies with incomplete or low signal-to-noise rotation curves, we retained 175 disk galaxies for model comparison. For computational feasibility, full Hamiltonian Monte Carlo (HMC) sampling was conducted for 39 representative galaxies spanning the range of stellar mass, surface brightness, and morphological type in the SPARC dataset[12]. These detailed posteriors provide well-converged constraints on key halo parameters such as the central density, scale radius, and inner slope.

For the remaining galaxies, we computed the Bayesian evidence ($\log Z$) and corresponding information criteria using approximate marginal-likelihood evaluation, ensuring a consistent basis for model selection across the entire sample. Each galaxy’s baryonic component was modeled as the quadrature sum of a gaseous disc, a stellar disc, and a stellar bulge (where applicable), scaled by a common stellar mass-to-light ratio. Four dark matter halo profiles were fitted to the data: the standard NFW, the generalized NFW (gNFW), the static Bose–Einstein Condensate (BEC), and its rotating variant (BEC-rot). Our analysis shows that while BEC-type profiles can reproduce cored density structures in low-mass systems, the majority of galaxies favor NFW or gNFW halos, indicating that classical cuspy profiles remain statistically preferred in the Bayesian comparison.

The present paper is organized as follows. We briefly present the Bose-Einstein Condensate and Navarro-Frenk-White model in Section 2, where the expressions of the rotational velocity and of the density distribution are obtained. The comparison of the theoretical model with the SPARC sample is performed in Section 3, where we present the fitting of a large number of rotation curves and calculate the evidence. We discuss and conclude our results in Section 4.

2 Background Theory

2.1 Formation and Properties of BEC Dark Matter

The Bose-Einstein Condensate suggests dark matter consists of ultralight bosonic particles (with masses billions of times lighter than an electron, e.g. $m_b \sim 10^{-22} \text{eV}$) that can coalesce into a macroscopic quantum state. On galactic scales, this means dark matter behaves not as a swarm of individual particles, but as a single, coherent wave spread across the entire galaxy.

For the condensation process to occur as a quantum mechanical correlation between boson particles, the mean interparticle distance need to become smaller than the de Broglie thermal wavelength, when the temperature of non-relativistic ideal Bose gas drops below the critical temperature

$$T_c = \frac{2\pi\hbar^2}{mk_B} \left(\frac{n}{\zeta(3/2)} \right)^{2/3}, \quad (1)$$

where n is the number density of boson gas, $\zeta(s)$ is the Riemann zeta function with $\zeta(3/2) \approx 2.612$, and k_B is Boltzmann's constant. This implies that the critical temperature is higher for lighter particles (small m) and denser environments (large n). This implies that ultralight bosons in the high-density early universe could easily form a galaxy-spanning BEC[8].

The dynamics of this giant quantum wave are governed by the Gross-Pitaevskii-Poisson (GPP) system[5]. In the mean-field limit, this key set of equations replaces the Newtonian dynamics of collisionless particles with the interplay between quantum pressure, gravity, and self-interactions, describing how this wave evolves:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + m\Phi\psi + u_0|\psi|^2\psi, \quad (2)$$

where the wave function $\psi(\mathbf{r}, t)$ gives the dark matter density through $\rho(\mathbf{r}, t) = m|\psi(\mathbf{r}, t)|^2$, $-\frac{\hbar^2}{2m} \nabla^2 \psi$ represents the quantum pressure that resists the compression of the wave function (and prevents the formation of a cuspy center), $m\Phi\psi$ is the gravitational pull from the total mass, and $u_0|\psi|^2\psi$ models the short-range self-interaction between bosons.

In the so-called Thomas-Fermi (TF) limit, the quantum pressure term is neglected, and the equilibrium state is a balance between pressure, gravity, and rotation, so we can often simplify the complex GPP system. For a non-rotating halo, this balance is Pressure Gradient + Gravity = 0. For slowly rotating BEC halos, the steady Euler equation that includes a centrifugal potential reduces to

$$\nabla P(\rho) + \rho \nabla \Phi + \rho \nabla \Phi_{\text{rot}} = \mathbf{0}, \quad (3)$$

where Φ is the gravitational potential satisfying Poisson's equation:

$$\nabla^2 \Phi = 4\pi G m \rho. \quad (4)$$

This indicates that the halo is supported against its own gravity partly by internal pressure and partly by rotation. The pressure P itself comes from the self-interaction of the BEC and has a specific form, $P \propto \rho^2$, which is crucial for predicting the halo's shape. For self-interacting BEC halos, the perturbative TF approach describes slow rotation as a correction to the non-rotating cored density profile. This fluid-like behavior contrasts with collisionless CDM halos, which use angular momentum distributions for rotation support [5, 13].

2.2 Dark Matter Self-interaction

A profound prediction of the Bose-Einstein Condensate model is the existence of a solitonic core—a stable, dense lump of quantum wave at the center of every dark matter halo. For self-gravitating ultralight bosons (BEC / fuzzy dark matter), the central solitonic core is supported against gravitational collapse by quantum pressure (or equivalently, the uncertainty principle). A simple scaling estimate may be obtained by balancing the total kinetic (quantum) energy against the gravitational potential energy.

Let m_b be the boson mass and $M_c = Nm_b$ the mass of the solitonic core with characteristic radius R_c . Using the uncertainty principle, the typical momentum of a particle confined within radius R_c is $p \sim \hbar/R_c$, so the kinetic energy of N particles is of order

$$E_{\text{kin}} \sim N E_{\text{kin}}^{(\text{per})} \sim \frac{N p^2}{2m_b} \sim \frac{N \hbar^2}{2m_b R_c^2} \sim \frac{\hbar^2 M_c}{2m_b^2 R_c^2}.$$

Equating the magnitudes $E_{\text{kin}} \sim E_{\text{grav}}$ yields the virial-like balance.

$$\frac{\hbar^2 M_c}{2m_b^2 R_c^2} \sim \frac{G M_c^2}{R_c}$$

A simple scaling shows that the core radius is inversely proportional to its mass:

$$R_c \sim \frac{\hbar^2}{G M_c m_b^2} \propto M_c^{-1}. \quad (5)$$

Thus, lighter halos form more extended cores. For instance, with $m_b = 10^{-22} \text{eV}/c^2$ and $M_c = 10^8 M_\odot$, one finds $R_c \sim \mathcal{O}(1) \text{ kpc}$, consistent with dwarf galaxy cores. [10].

The presence of a dark matter halo becomes unambiguous when rotation curves remain flat or rising at large radii $R \gtrsim 5R_0$, where the gravitational influence of baryonic matter alone would predict declining velocities.

2.3 Spherical Symmetry and Asymmetric Disc Model: How We Measure Dark Matter

To understand how to test dark matter models, we first need to understand how astronomers measure mass in galaxies. The key tool is the rotation curve—a

plot showing how fast stars and gas orbit at different distances from the galactic center.

For a spherical mass distribution that has a spherically symmetric gravitational potential (like a simple dark matter halo), the circular velocity v_{circ} of orbiting objects is related to the mass enclosed within radius R by

$$v_{\text{circ}}^2 = \frac{GM(R)}{R} \quad (6)$$

where $M(R)$ is the total mass within radius R , and G is the gravitational constant.

Most galaxies we study are disk-shaped, making calculations of the gravitational force in a disk more complex because mass at all radii contributes to the force at any given point, such that the expression of the rotation velocity for a thin disk is:

$$v^2(R) = 2\pi G \int_0^\infty \mathcal{K}(R, R') \Sigma(R') dR' \quad (7)$$

where $\Sigma(R')$ is the mass density at radius R' , and $\mathcal{K}$ is a kernel function that accounts for gravitational contributions from all parts of the disk.

Note that old stars cannot have circular rotational curves simulated by this expression due to the increasing dispersion of bulk rotational velocity. Therefore, interpretation of a flat rotation curve, based on the assumption that dark matter is spheroidally distributed in a 'dark halo', gives information about the total mass of the disc galaxy.

The SPARC database provides these crucial rotation curves along with detailed measurements of the visible mass components, allowing us to isolate the gravitational contribution of dark matter and test different halo models against the data.

2.4 Bose-Einstein-Condensate (BEC) Profile

Rotating Bose-Einstein Condensate (BEC) Profile. The BEC model assumes that dark matter consists of ultralight bosons forming a coherent condensate on galactic scales. In the Thomas-Fermi limit, the density is modeled using solutions to the Gross-Pitaevskii-Poisson equations. We adopt the rotating Bose-Einstein condensate model derived in [14], which provides an analytical expression for the dark matter rotation curve::

$$v_{\text{DM, BEC}}^2(r) = 4\pi G \rho_c R^2 \left[\frac{\sin(\pi r/R)}{\pi r/R} - \cos(\pi r/R) \right] \quad (8)$$

with central density ρ_0 and cutoff radius R_c . This is a *phenomenological* profile: it is motivated by condensate physics but used primarily as an effective model to fit rotation curves.

Non-rotating BEC Soliton Profile. From solutions of the Schrödinger-Poisson system for ultralight

scalar dark matter ("fuzzy" DM), one obtains a solitonic core density profile:

$$\rho_{\text{soliton}}(r) = \frac{\rho_c}{\left[1 + 0.091 (r/r_c)^2\right]^8},$$

where ρ_c is the central density and r_c the core radius. This arises from the ground-state wavefunction of the condensate and is distinct from the phenomenological sinusoidal BEC profile above[19].

2.5 Navarro-Frenk-White (NFW) Halo Profile

The Navarro-Frenk-White (NFW) profile arises naturally from cold dark matter (CDM) cosmological simulations, assuming dark matter is made of heavy, collisionless particles that responds to only gravity by being pulled into a 'cuspy' structure. NFW models dark matter halos with two parameters: a scale radius r_s , and a characteristic volume density ρ_s . For convenience, the free parameters in our fits are the concentration C_{200} and the rotational velocity V_{200} , defined as

$$C_{200} = r_{200}/r_s; V_{200} = 10C_{200}r_s H_0,$$

r_{200} is the radius inside of which the average halo density is 200 times the critical density of the universe. Halos are nearly self-similar across masses and epochs, with larger halos being less dense. This size-density relation, tied to formation time, offers a means to constrain cosmological parameters and understand structure formation.

Standardized NFW Profile As the popular density profile with ρ inversely proportional to r has the problem of mass being infinite, and its more practical modification cannot give flat rotation curves, we finally select an N-body DM-only simulations of structure formation that predict a cuspy density distribution profile[15]:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2},$$

where ρ_0 is the central density and r_s is the scale radius, $\rho \propto r^{-1}$ at small radii and $\rho \propto r^{-3}$ at large radii. The corresponding circular velocity is

$$V_{\text{NFW}}^2(r) = \frac{GM(r)}{r}, \quad (9)$$

$$M(r) = 4\pi \rho_s r_s^3 \left[\ln\left(1 + \frac{r}{r_s}\right) - \frac{r/r_s}{1 + r/r_s} \right]. \quad (10)$$

Generalized NFW (gNFW) Profile. The gNFW profile is a more flexible extension of the standardized NFW profile. It introduces an adjustable inner slope parameter γ , which describes the center as a steep 'cusp' ($\gamma \approx 1$) or a flat 'core' ($\gamma \approx 0$), so that

$$\rho_{\text{gNFW}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)^\gamma \left(1 + \frac{r}{r_s}\right)^{3-\gamma}}.$$

Again, $\gamma = 1$ one recovers the standard NFW form, while $\gamma < 1$ produces a shallower, cored profile[21].

3 Methodology

In the present Section, we perform a Bayesian comparison of four dark matter models—NFW, generalized NFW, and rotating/non-rotating BEC profiles—using SPARC rotation curves. Our analysis employs Markov Chain Monte Carlo (MCMC) sampling to fit each model and computes Bayesian evidence for rigorous model selection. This framework extracts halo parameters (density, radius, inner slope) while quantifying uncertainties, determining which model best describes galactic dynamics.

3.1 The SPARC Sample

The primary dataset used in this study is the Spitzer Photometry and Accurate Rotation Curves (SPARC) database [12], which provides high-quality rotation curves and $3.6 \mu\text{m}$ photometry for 175 disk galaxies. The sample spans a wide range of luminosities, stellar masses, surface brightnesses, gas fractions, and morphological types, making it ideal for testing dark matter halo models.

Each SPARC galaxy entry includes radius values (kpc) and observed rotational velocities V_{obs} from extended HI/H α measurements; pre-computed baryonic velocity contributions: $V_{\text{disk}}, V_{\text{bulge}}, V_{\text{gas}}$; near-infrared ($3.6 \mu\text{m}$) surface photometry with minimal scatter in stellar mass-to-light ratio ($\Upsilon_{\star} \sim 0.5\text{--}0.7$); nuisance parameters: galaxy distance (from multiple estimators) and disk inclination

The $3.6 \mu\text{m}$ wavelength is optimal for stellar mass estimation because emission is dominated by older, low-mass stars that trace the bulk of stellar mass, thereby reducing uncertainties and alleviating the disk-halo degeneracy.

In SPARC, the stellar disk and bulge velocity contributions are tabulated under the assumption of a unit mass-to-light ratio ($\Upsilon = 1$). These tabulated values, denoted with subscript 'tab', must be rescaled to obtain realistic stellar contributions:

$$V_{\text{disk}}(r) = \sqrt{\Upsilon_{\text{disk}}} V_{\text{disk,tab}}(r), \quad (11)$$

$$V_{\text{bulge}}(r) = \sqrt{\Upsilon_{\text{bulge}}} V_{\text{bulge,tab}}(r). \quad (12)$$

The gas component $V_{\text{gas}}(r) = V_{\text{gas,tab}}(r)$ is independent of the stellar mass-to-light ratio, as gas masses are derived directly from HI surface density profiles.

For simplicity, we assume a common stellar mass-to-light ratio $\Upsilon_{\star}$ for both disk and bulge components. The total modeled rotation curve is then the quadrature sum of all components:

$$V_{\text{model}}^2(r) = V_{\text{gas}}^2(r) \quad (13)$$

$$+ \Upsilon_{\star} [V_{\text{disk,tab}}^2(r) + V_{\text{bulge,tab}}^2(r)] + V_{\text{halo}}^2(r). \quad (14)$$

This scaling arises from the relation $V^2(r) \propto M(r)$ and $M(r) \propto \Upsilon$, which implies $V(r) \propto \sqrt{\Upsilon}$ [4].

This comprehensive dataset enables detailed decomposition of galaxy rotation curves into baryonic and dark matter components, making it ideally suited for testing alternative halo models, including NFW, generalized NFW (gNFW), and Bose-Einstein condensate (BEC) profiles.

3.2 Computational Framework and Implementation

Our analysis extends the open-source rotation curve fitting framework developed by [1], implementing a comprehensive Bayesian inference pipeline for dark matter model comparison.

In our approach, the theoretical rotation curve models are fitted to the SPARC data using a multi-step optimization procedure. For each galaxy in the sample, we determine the optimal parameters of the dark matter halo by minimizing the discrepancy between the observed rotation curve $V_{\text{obs}}(r)$ and the theoretical prediction $V_{\text{model}}(r)$. We extend the `mlpq` codebase to implement BEC halo profiles as custom likelihood functions, add a generalized NFW profile with variable inner slope, compute the Bayesian evidence for model comparison, and then automate the processing of all SPARC galaxies.

3.2.1 Bayesian Inference Framework

Defining the Likelihood Function for each galaxy and model $\mathcal{M}$, we get the likelihood:

$$\mathcal{L}(\theta|\mathbf{V}_{\text{obs}}) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left[-\frac{(V_{\text{obs},i} - V_{\text{model}}(r_i; \theta))^2}{2\sigma_i^2} \right] \quad (15)$$

where the total rotation curve is:

$$V_{\text{model}}^2(r) = V_{\text{gas}}^2(r) + \Upsilon_{\star} V_{\star}^2(r) + V_{\text{DM}}^2(r; \theta_{\text{halo}})$$

Model parameters θ are assigned as follows:

Halo Profile	θ
NFW	$\{\Upsilon_{\star}, \rho_s, r_s\}$
gNFW	$\{\Upsilon_{\star}, \rho_s, r_s, \gamma\}$
BEC	$\{\Upsilon_{\star}, \rho_c, R\}$
BEC-rot	$\{\Upsilon_{\star}, \rho_c, R, \Omega\}$

Table 1: model parameters θ

For the prior distributions, we use physically motivated priors, with the Stellar mass-to-light ratio $\log \Upsilon_{\star} \sim \mathcal{N}(\log(0.5), 0.1^2)$, BEC central density $\log \rho_c \sim \mathcal{N}(6, 2^2)$ in $\log(M_{\odot}/\text{kpc}^3)$, BEC halo radius $\log R \sim \mathcal{N}(1, 1^2)$ in $\log(\text{kpc})$ in kpc, NFW concentration $\log c \sim \mathcal{N}(\log c_{\text{cosmo}}, 0.1^2)$.

The sampling procedure is carried out using the No-U-Turn Sampler (NUTS) in NumPyro with 4 parallel chains with 1000 warmup + 2000 sampling steps, a convergence criteria with $\hat{R} < 1.01$, ESS > 400, and Gelman-Rubin statistic and trace plots for diagnostics[17].

3.3 Model Comparison Framework

3.3.1 Bayesian Evidence and Information Criteria

We compute marginal likelihood using the harmonic mean estimator:

$$Z = P(\mathbf{V}_{\text{obs}}|\mathcal{M}) = \int \mathcal{L}(\theta|\mathbf{V}_{\text{obs}})P(\theta)d\theta \quad (16)$$

$$\approx \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{\mathcal{L}(\theta_i|\mathbf{V}_{\text{obs}})} \right]^{-1} \quad (17)$$

We employ three complementary metrics: **Bayesian Evidence** (Z), the direct probability of data given model; **Bayes Factors**, with $BF_{ij} = Z_i/Z_j$, interpreted via Jeffrey's scale; **Modified BIC**, where $\text{BIC} = k \ln N - 2 \ln \hat{P}$, $\hat{P}$ is the maximum posterior density[11, 20]. Note that the harmonic-mean estimator can be noisy, so we stabilized it with a log-sum-exp trick.

4 Results and Model Comparison

4.1 Automated Workflow

Our workflow consists of loading SPARC data with quality cuts ($\text{SNR} > 3$, $R_{\text{max}} > 5R_0$), running MCMC for all 4 models in parallel for all galaxies, computing the Bayesian evidence and model comparison metrics, generating diagnostic plots and summary tables, then finally aggregating the results across the full sample.

First testing with galaxy DDO154, we ran NFW, gNFW, BEC, BECrot as the four different models for all galaxies. For each model, MCMC sampled the posterior distribution of the model parameters, while 500 warm-ups and 1000 samples gave enough samples to estimate means, stds, and credible intervals. In the code, n_{eff} tells us effective sample size, r_{hat} tells us convergence of chains ($r_{\text{hat}} \approx 1$ is good; larger values mean poor convergence). "Number of divergences: 0" means our sampler was stable (good).

The output table here is posterior summary statistics: mean, std, median, 5% – 95% interval.

To illustrate the qualitative differences between models, we present rotation curve decompositions for the selected 36 galaxies spanning the mass and morphology range of our sample. For computational feasibility, we ran full posterior sampling for only these 36 representative galaxies, selected to span the mass, morphology, and surface brightness range of the SPARC sample. For the remaining galaxies, Bayesian evidence was computed using approximate marginal likelihood evaluation.

Table 2: MCMC posterior summaries for 36 galaxy models.

Galaxy	Model	Parameter	Mean	Std	Median	5%	95%
DDO154	NFW	$\log Y$	-1.5235	0.2257	-1.5308	-1.7382	-1.3006
DDO154	NFW	$\log r_s$	2.6711	0.1	2.6691	2.5691	2.7763
DDO154	NFW	$\log \rho_s$	14.2899	0.1275	14.2899	14.1578	14.4198
DDO154	gNFW	γ	0.6861	0.3689	0.3958	0.3811	1.0986
DDO154	gNFW	$\log Y$	-2.5628	1.0603	-2.8767	-3.6171	-1.4086
DDO154	gNFW	$\log r_s$	4.2448	1.5687	4.4087	2.6354	5.8157
DDO154	gNFW	$\log \rho_s$	12.576	1.7075	12.3698	10.8678	14.335
DDO154	BEC	$\log R$	1.6698	0.0106	1.6693	1.6597	1.6802
DDO154	BEC	$\log Y$	0.3309	0.0968	0.3371	0.2365	0.4265
DDO154	BEC	$\log \rho_c$	13.9817	0.0175	13.982	13.9646	13.9982
DDO154	BEC_rot	$\log R$	1.6698	0.0103	1.6696	1.6592	1.68
DDO154	BEC_rot	$\log Y$	0.3317	0.0946	0.3349	0.2385	0.4245
DDO154	BEC_rot	$\log \omega$	-4.6059	0.4905	-4.6016	-5.1115	-4.104
DDO154	BEC_rot	$\log \rho_c$	13.9817	0.0171	13.9819	13.9645	13.9988
ESO444	NFW	$\log Y$	-0.741	0.301	-0.734	-1.250	-0.246
ESO444	NFW	$\log r_s$	2.402	0.292	2.388	1.936	2.901
ESO444	NFW	$\log \rho_s$	15.462	0.369	15.467	14.843	16.060
ESO444	gNFW	γ	0.994	0.296	1.004	0.494	1.467
ESO444	gNFW	$\log Y$	-0.738	0.288	-0.742	-1.198	-0.261
ESO444	gNFW	$\log r_s$	2.415	0.292	2.394	1.962	2.926
ESO444	gNFW	$\log \rho_s$	15.446	0.365	15.468	14.821	16.023
ESO444	BEC	$\log R$	1.305	0.038	1.302	1.246	1.372
ESO444	BEC	$\log Y$	0.064	0.358	0.074	-0.551	0.619
ESO444	BEC	$\log \rho_c$	15.383	0.081	15.387	15.240	15.509
ESO444	BEC_rot	$\log R$	1.303	0.038	1.299	1.248	1.373
ESO444	BEC_rot	$\log Y$	0.051	0.356	0.058	-0.571	0.605
ESO444	BEC_rot	$\log \omega$	-4.622	0.491	-4.621	-5.427	-3.786
ESO444	BEC_rot	$\log \rho_c$	15.387	0.080	15.391	15.250	15.509
NGC0289	NFW	$\log Y$	-0.6241	0.1301	-0.6131	-0.7563	-0.4922
NGC0289	NFW	$\log r_s$	3.022	0.1858	3.0255	2.8335	3.206
NGC0289	NFW	$\log \rho_s$	15.3134	0.3832	15.3002	14.9327	15.705
NGC0289	gNFW	γ	0.998	0.292	0.9995	0.7111	1.289
NGC0289	gNFW	$\log Y$	-0.6143	0.1199	-0.6036	-0.7321	-0.4953
NGC0289	gNFW	$\log r_s$	3.028	0.1727	3.0288	2.8555	3.2037
NGC0289	gNFW	$\log \rho_s$	15.2984	0.3557	15.2937	14.9356	15.6551
NGC0289	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC0289	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC0289	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC0289	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC0289	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC0289	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC0289	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894
NGC0801	NFW	$\log Y$	-0.5613	0.0382	-0.5597	-0.599	-0.5245
NGC0801	NFW	$\log r_s$	3.745	0.2468	3.7487	3.4936	3.9875
NGC0801	NFW	$\log \rho_s$	14.0261	0.4419	14.0021	13.5975	14.4718
NGC0801	gNFW	γ	1.0058	0.2994	1.0049	0.71	1.3073
NGC0801	gNFW	$\log Y$	-0.5624	0.0393	-0.5596	-0.6005	-0.5228
NGC0801	gNFW	$\log r_s$	3.7522	0.2552	3.7568	3.5043	4.0107
NGC0801	gNFW	$\log \rho_s$	14.016	0.4585	13.9974	13.5579	14.4593
NGC0801	BEC	$\log R$	1.261	0.0033	1.2611	1.2576	1.2644
NGC0801	BEC	$\log Y$	-1.5936	0.1336	-1.5842	-1.7279	-1.4616
NGC0801	BEC	$\log \rho_c$	17.8067	0.0412	17.8054	17.7661	17.8481
NGC0801	BEC_rot	$\log R$	1.2612	0.0033	1.2612	1.2577	1.2647
NGC0801	BEC_rot	$\log Y$	-1.6003	0.1338	-1.5902	-1.73	-1.4682
NGC0801	BEC_rot	$\log \omega$	-4.6158	0.4948	-4.6253	-5.1041	-4.103
NGC0801	BEC_rot	$\log \rho_c$	17.8085	0.0399	17.808	17.769	17.8478
NGC1003	NFW	$\log Y$	-0.472	0.0924	-0.4676	-0.5678	-0.3804
NGC1003	NFW	$\log r_s$	3.5601	0.1188	3.558	3.4397	3.6809
NGC1003	NFW	$\log \rho_s$	13.7178	0.1885	13.7209	13.5242	13.9096
NGC1003	gNFW	γ	1.0059	0.3015	1.0181	0.7071	1.3034
NGC1003	gNFW	$\log Y$	-0.4665	0.0895	-0.4592	-0.5543	-0.3807
NGC1003	gNFW	$\log r_s$	3.5662	0.119	3.5689	3.4484	3.6797
NGC1003	gNFW	$\log \rho_s$	13.7081	0.1882	13.7034	13.5299	13.8921
NGC1003	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC1003	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC1003	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC1003	BEC_rot	$\log R$	0.081	0.962	0.0817	-0.8812	1.043
NGC1003	BEC_rot	$\log Y$	-0.5019	0.6511	-0.0091	-1.1837	0.1397
NGC1003	BEC_rot	$\log \omega$	1.0528	1.022	1.1384	0.9303	1.1529
NGC1003	BEC_rot	$\log \rho_c$	10.4222	9.9329	10.209	0.4894	20.3751
NGC1705	NFW	$\log Y$	-0.0623	0.3121	-0.0315	-0.356	0.2499
NGC1705	NFW	$\log r_s$	1.3496	0.3532	1.3286	1.0068	1.6959
NGC1705	NFW	$\log \rho_s$	17.2246	0.6162	17.2474	16.5993	17.8435
NGC1705	gNFW	γ	1.0036	0.3061	0.9987	0.6969	1.306
NGC1705	gNFW	$\log Y$	-0.0526	0.3065	-0.0287	-0.3556	0.2473
NGC1705	gNFW	$\log r_s$	1.357	0.3523	1.3517	1.004	1.7087
NGC1705	gNFW	$\log \rho_s$	17.2107	0.6136	17.2	16.5858	17.8261
NGC1705	BEC	$\log R$	1.6262	0.0672	1.6158	1.5626	1.6896
NGC1705	BEC	$\log Y$	0.7408	0.1119	0.7442	0.6351	0.8492
NGC1705	BEC	$\log \rho_c$	14.9143	0.1494	14.9278	14.7651	15.0648
NGC1705	BEC_rot	$\log R$	1.6206	0.0663	1.6119	1.5579	1.6816
NGC1705	BEC_rot	$\log Y$	0.7323	0.1102	0.7346	0.624	0.8383
NGC1705	BEC_rot	$\log \omega$	-4.6139	0.5078	-4.6099	-5.1218	-4.1016
NGC1705	BEC_rot	$\log \rho_c$	14.9281	0.1469	14.9384	14.7859	15.0736

Table 3: MCMC posterior summaries for 36 galaxy models.

Galaxy	Model	Parameter	Mean	Std	Median	5%	95%
NGC2403	NFW	$\log Y$	0.3159	0.3181	0.3165	-0.0036	0.6346
NGC2403	NFW	$\log r_s$	5.9802	0.2027	5.9817	5.7765	6.1829
NGC2403	NFW	$\log \rho_s$	9.9752	1.6246	9.9797	8.3473	11.5997
NGC2403	gNFW	γ	0.562	0.1855	0.5628	0.376	0.7482
NGC2403	gNFW	$\log Y$	0.5504	0.0674	0.5649	0.472	0.617
NGC2403	gNFW	$\log r_s$	6.2006	0.031	6.2006	6.1696	6.2315
NGC2403	gNFW	$\log \rho_s$	9.5344	0.9379	9.5399	8.5952	10.4722
NGC2403	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC2403	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC2403	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC2403	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC2403	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC2403	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC2403	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894
NGC2683	NFW	$\log Y$	-0.9814	0.1067	-0.9721	-1.0863	-0.8784
NGC2683	NFW	$\log r_s$	2.2656	0.272	2.2598	1.9953	2.5319
NGC2683	NFW	$\log \rho_s$	16.6638	0.5876	16.6609	16.0715	17.2539
NGC2683	gNFW	γ	1.006	0.2961	1.0073	0.7041	1.3044
NGC2683	gNFW	$\log Y$	-0.9836	0.1133	-0.9713	-1.0874	-0.8716
NGC2683	gNFW	$\log r_s$	2.2524	0.281	2.2522	1.9789	2.5253
NGC2683	gNFW	$\log \rho_s$	16.69	0.6126	16.6842	16.0853	17.2924
NGC2683	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC2683	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC2683	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC2683	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC2683	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC2683	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC2683	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894
NGC2841	NFW	$\log Y$	-0.7074	0.0315	-0.7059	-0.7375	-0.6762
NGC2841	NFW	$\log r_s$	3.4824	0.1195	3.4816	3.3729	3.5984
NGC2841	NFW	$\log \rho_s$	15.3898	0.2247	15.3891	15.1697	15.5965
NGC2841	gNFW	γ	0.9988	0.3024	0.9917	0.6944	1.2996
NGC2841	gNFW	$\log Y$	-0.701	0.0286	-0.6999	-0.7302	-0.6728
NGC2841	gNFW	$\log r_s$	3.5079	0.1135	3.508	3.4018	3.6154
NGC2841	gNFW	$\log \rho_s$	15.3419	0.2116	15.3392	15.1369	15.5373
NGC2841	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC2841	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC2841	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC2841	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC2841	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC2841	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC2841	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894
NGC2903	NFW	$\log Y$	-1.3863	0.0939	-1.3797	-1.47	-1.2944
NGC2903	NFW	$\log r_s$	1.6873	0.0984	1.6873	1.5963	1.7867
NGC2903	NFW	$\log \rho_s$	18.274	0.2217	18.2724	18.0474	18.482
NGC2903	gNFW	γ	1.0063	0.2997	1.0026	0.7037	1.2986
NGC2903	gNFW	$\log Y$	-1.3664	0.0842	-1.3648	-1.4497	-1.2841
NGC2903	gNFW	$\log r_s$	1.707	0.0955	1.7057	1.6129	1.7969
NGC2903	gNFW	$\log \rho_s$	18.2289	0.2129	18.2299	18.0263	18.44
NGC2903	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC2903	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC2903	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC2903	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC2903	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC2903	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC2903	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894
NGC2915	NFW	$\log Y$	-0.997	0.251	-0.985	-1.435	-0.593
NGC2915	NFW	$\log r_s$	1.794	0.190	1.792	1.497	2.126
NGC2915	NFW	$\log \rho_s$	16.620	0.330	16.624	16.059	17.141
NGC2915	gNFW	γ	0.976	0.294	0.974	0.492	1.476
NGC2915	gNFW	$\log Y$	-0.994	0.261	-0.996	-1.409	-0.572
NGC2915	gNFW	$\log r_s$	1.793	0.194	1.791	1.474	2.123
NGC2915	gNFW	$\log \rho_s$	16.622	0.340	16.615	16.063	17.188
NGC2915	BEC	$\log R$	0.420	1.721	0.385	-1.301	2.173
NGC2915	BEC	$\log Y$	-0.637	0.413	-0.978	-1.018	0.033
NGC2915	BEC	$\log \rho_c$	7.972	6.443	7.801	1.528	14.508
NGC2915	BEC_rot	$\log R$	2.140	0.026	2.138	2.100	2.185
NGC2915	BEC_rot	$\log Y$	-0.273	0.237	-0.258	-0.677	0.099
NGC2915	BEC_rot	$\log \omega$	-4.596	0.507	-4.591	-5.405	-3.749
NGC2915	BEC_rot	$\log \rho_c$	14.417	0.076	14.419	14.293	14.538
NGC2998	NFW	$\log Y$	-1.0593	0.2634	-1.0565	-1.3295	-0.7898
NGC2998	NFW	$\log r_s$	2.4789	0.2342	2.4421	2.2522	2.729
NGC2998	NFW	$\log \rho_s$	16.8043	0.5367	16.8794	16.2241	17.3331
NGC2998	gNFW	γ	1.0039	0.3081	1.0024	0.6977	1.3006
NGC2998	gNFW	$\log Y$	-1.0331	0.2566	-1.0417	-1.2935	-0.7627
NGC2998	gNFW	$\log r_s$	2.5024	0.2565	2.4564	2.2696	2.7636
NGC2998	gNFW	$\log \rho_s$	16.7511	0.5797	16.8575	16.1508	17.2822
NGC2998	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC2998	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC2998	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC2998	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC2998	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC2998	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC2998	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894

Table 4: MCMC posterior summaries for 36 galaxy models.

Galaxy	Model	Parameter	Mean	Std	Median	5%	95%
NGC3198	NFW	$\log Y$	-0.8144	0.0803	-0.8104	-0.8927	-0.7338
NGC3198	NFW	$\log r_s$	2.6628	0.0818	2.6612	2.5824	2.749
NGC3198	NFW	$\log \rho_s$	15.8315	0.1727	15.8347	15.6512	16.0016
NGC3198	gNFW	γ	0.9955	0.2974	0.9848	0.7146	1.2969
NGC3198	gNFW	$\log Y$	-0.805	0.0813	-0.8015	-0.886	-0.7245
NGC3198	gNFW	$\log r_s$	2.6724	0.0854	2.6681	2.5876	2.757
NGC3198	gNFW	$\log \rho_s$	15.8111	0.1799	15.8173	15.634	15.9852
NGC3198	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC3198	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC3198	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC3198	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC3198	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC3198	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC3198	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894
NGC3741	NFW	$\log Y$	-0.829	0.258	-0.811	-1.273	-0.419
NGC3741	NFW	$\log r_s$	2.790	0.222	2.782	2.435	3.164
NGC3741	NFW	$\log \rho_s$	14.085	0.281	14.091	13.605	14.537
NGC3741	gNFW	γ	1.002	0.297	1.010	0.526	1.495
NGC3741	gNFW	$\log Y$	-0.814	0.268	-0.811	-1.251	-0.401
NGC3741	gNFW	$\log r_s$	2.795	0.231	2.780	2.442	3.220
NGC3741	gNFW	$\log \rho_s$	14.078	0.291	14.094	13.556	14.528
NGC3741	BEC	$\log R$	1.843	0.030	1.841	1.796	1.896
NGC3741	BEC	$\log Y$	0.435	0.170	0.443	0.145	0.690
NGC3741	BEC	$\log \rho_c$	13.868	0.058	13.870	13.767	13.962
NGC3741	BEC_rot	$\log R$	1.844	0.030	1.842	1.798	1.896
NGC3741	BEC_rot	$\log Y$	0.442	0.159	0.454	0.158	0.683
NGC3741	BEC_rot	$\log \omega$	-4.604	0.487	-4.599	-5.411	-3.813
NGC3769	NFW	$\log Y$	-1.0855	0.1824	-1.0749	-1.2737	-0.9053
NGC3769	NFW	$\log r_s$	2.3978	0.2128	2.4000	2.1886	2.6061
NGC3769	NFW	$\log \rho_s$	15.9583	0.3986	15.9563	15.5693	16.3455
NGC3769	gNFW	γ	0.9889	0.3064	0.9919	0.6848	1.2902
NGC3769	gNFW	$\log Y$	-1.0558	0.1834	-1.0495	-1.2265	-0.8712
NGC3769	gNFW	$\log r_s$	2.4269	0.2159	2.4225	2.2106	2.6451
NGC3769	gNFW	$\log \rho_s$	15.9026	0.4040	15.9024	15.4847	16.3155
NGC3769	BEC	$\log R$	-0.171	1.1303	-0.171	-1.3014	0.9593
NGC3769	BEC	$\log Y$	-0.7292	0.289	-0.7292	-1.0182	-0.4401
NGC3769	BEC	$\log \rho_c$	0.4444	1.084	0.4444	-0.6396	1.5284
NGC3769	BEC_rot	$\log R$	1.4364	0.3933	1.4364	1.043	1.8297
NGC3769	BEC_rot	$\log Y$	-0.3056	0.4453	-0.3056	-0.7509	0.1397
NGC3769	BEC_rot	$\log \omega$	-0.3793	1.5322	-0.3793	-1.9115	1.1529
NGC3769	BEC_rot	$\log \rho_c$	0.397	0.0924	0.397	0.3046	0.4894
NGC4138	NFW	$\log Y$	-0.962	0.122	-0.953	-1.175	-0.770
NGC4138	NFW	$\log r_s$	2.263	0.344	2.261	1.713	2.847
NGC4138	NFW	$\log \rho_s$	16.318	0.670	16.311	15.172	17.410
NGC4138	gNFW	γ	0.990	0.308	0.989	0.497	1.511
NGC4138	gNFW	$\log Y$	-0.953	0.124	-0.944	-1.181	-0.761
NGC4138	gNFW	$\log r_s$	2.290	0.359	2.271	1.704	2.905
NGC4138	gNFW	$\log \rho_s$	16.256	0.693	16.255	15.143	17.378
NGC4138	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
NGC4138	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
NGC4138	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
NGC4138	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
NGC4138	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
NGC4138	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
NGC4138	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
NGC4214	NFW	$\log Y$	-0.352	0.229	-0.326	-0.780	-0.023
NGC4214	NFW	$\log r_s$	1.969	0.400	1.952	1.337	2.652
NGC4214	NFW	$\log \rho_s$	16.381	0.618	16.384	15.377	17.413
NGC4214	gNFW	γ	1.009	0.306	1.016	0.501	1.500
NGC4214	gNFW	$\log Y$	-0.358	0.239	-0.344	-0.781	-0.014
NGC4214	gNFW	$\log r_s$	1.981	0.396	1.982	1.338	2.675
NGC4214	gNFW	$\log \rho_s$	16.364	0.612	16.334	15.356	17.405
NGC4214	BEC	$\log R$	1.718	0.142	1.693	1.558	1.959
NGC4214	BEC	$\log Y$	0.182	0.121	0.186	-0.015	0.373
NGC4214	BEC	$\log \rho_c$	14.955	0.208	14.978	14.575	15.265
NGC4214	BEC_rot	$\log R$	1.719	0.133	1.694	1.556	1.976
NGC4214	BEC_rot	$\log Y$	0.179	0.117	0.182	-0.020	0.361
NGC4214	BEC_rot	$\log \omega$	-4.605	0.491	-4.600	-5.396	-3.806
NGC4214	BEC_rot	$\log \rho_c$	14.956	0.208	14.967	14.583	15.266
NGC5033	NFW	$\log Y$	-1.699	0.105	-1.692	-1.875	-1.540
NGC5033	NFW	$\log r_s$	2.137	0.084	2.136	1.998	2.271
NGC5033	NFW	$\log \rho_s$	17.584	0.194	17.585	17.270	17.912
NGC5033	gNFW	γ	1.000	0.307	0.993	0.514	1.505
NGC5033	gNFW	$\log Y$	-1.685	0.099	-1.680	-1.854	-1.528
NGC5033	gNFW	$\log r_s$	2.148	0.082	2.146	2.010	2.286
NGC5033	gNFW	$\log \rho_s$	17.558	0.187	17.565	17.241	17.873
NGC5033	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
NGC5033	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
NGC5033	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
NGC5033	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
NGC5033	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
NGC5033	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
NGC5033	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489

Table 5: MCMC posterior summaries for 36 galaxy models.

Galaxy	Model	Parameter	Mean	Std	Median	5%	95%
NGC5055	NFW	$\log Y$	-0.942	0.268	-0.899	-1.254	-0.672
NGC5055	NFW	$\log r_s$	4.849	2.333	4.916	2.447	7.183
NGC5055	NFW	$\log \rho_s$	13.115	3.403	12.963	9.712	16.672
NGC5055	gNFW	γ	1.013	0.209	1.022	0.639	1.405
NGC5055	gNFW	$\log Y$	-0.960	0.249	-0.959	-1.249	-0.692
NGC5055	gNFW	$\log r_s$	4.632	2.116	4.698	2.453	6.760
NGC5055	gNFW	$\log \rho_s$	13.388	3.129	13.232	10.249	16.659
NGC5055	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
NGC5055	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
NGC5055	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
NGC5055	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
NGC5055	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
NGC5055	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
NGC5055	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
NGC6015	NFW	$\log Y$	-0.661	0.113	-0.669	-0.820	-0.504
NGC6015	NFW	$\log r_s$	2.131	0.196	2.101	1.971	2.308
NGC6015	NFW	$\log \rho_s$	16.872	0.389	16.929	16.486	17.230
NGC6015	gNFW	γ	1.010	0.295	1.018	0.517	1.489
NGC6015	gNFW	$\log Y$	-0.654	0.108	-0.658	-0.814	-0.484
NGC6015	gNFW	$\log r_s$	2.135	0.179	2.106	1.966	2.335
NGC6015	gNFW	$\log \rho_s$	16.862	0.356	16.915	16.416	17.234
NGC6015	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
NGC6015	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
NGC6015	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
NGC6015	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
NGC6015	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
NGC6015	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
NGC6015	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
NGC6503	NFW	$\log Y$	-0.938	0.071	-0.936	-1.056	-0.824
NGC6503	NFW	$\log r_s$	2.044	0.078	2.045	1.921	2.171
NGC6503	NFW	$\log \rho_s$	16.598	0.159	16.596	16.337	16.848
NGC6503	gNFW	γ	1.112	0.241	1.234	0.629	1.375
NGC6503	gNFW	$\log Y$	-1.109	0.189	-1.208	-1.296	-0.832
NGC6503	gNFW	$\log r_s$	4.195	2.139	4.310	1.953	6.337
NGC6503	gNFW	$\log \rho_s$	13.790	2.785	13.556	11.006	16.786
NGC6503	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
NGC6503	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
NGC6503	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
NGC6503	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
NGC6503	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
NGC6503	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
NGC6503	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
NGC6674	NFW	$\log Y$	-0.365	0.037	-0.362	-0.432	-0.306
NGC6674	NFW	$\log r_s$	4.720	0.214	4.714	4.361	5.079
NGC6674	NFW	$\log \rho_s$	12.821	0.328	12.823	12.290	13.394
NGC6674	gNFW	γ	1.008	0.298	1.004	0.533	1.508
NGC6674	gNFW	$\log Y$	-0.361	0.035	-0.357	-0.422	-0.309
NGC6674	gNFW	$\log r_s$	4.749	0.212	4.749	4.416	5.092
NGC6674	gNFW	$\log \rho_s$	12.778	0.321	12.771	12.269	13.295
NGC6674	BEC	$\log R$	-0.833	0.650	-0.832	-1.483	-0.182
NGC6674	BEC	$\log Y$	-0.422	0.787	-0.309	-1.301	0.371
NGC6674	BEC	$\log \rho_c$	10.459	10.481	10.461	-0.083	20.965
NGC6674	BEC_rot	$\log R$	-0.606	0.424	-0.606	-1.032	-0.181
NGC6674	BEC_rot	$\log Y$	-0.300	0.296	-0.189	-0.714	0.002
NGC6674	BEC_rot	$\log \omega$	0.894	0.210	1.021	0.564	1.103
NGC6674	BEC_rot	$\log \rho_c$	15.009	6.060	17.382	3.368	20.592
NGC7331	NFW	$\log Y$	-1.450	0.037	-1.449	-1.512	-1.390
NGC7331	NFW	$\log r_s$	3.353	0.143	3.349	3.124	3.597
NGC7331	NFW	$\log \rho_s$	15.345	0.247	15.348	14.924	15.745
NGC7331	gNFW	γ	1.005	0.302	1.005	0.521	1.511
NGC7331	gNFW	$\log Y$	-1.445	0.037	-1.444	-1.507	-1.386
NGC7331	gNFW	$\log r_s$	3.376	0.150	3.372	3.135	3.629
NGC7331	gNFW	$\log \rho_s$	15.306	0.259	15.307	14.874	15.728
NGC7331	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
NGC7331	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
NGC7331	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
NGC7331	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
NGC7331	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
NGC7331	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
NGC7331	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC02487	NFW	$\log Y$	-0.950	0.135	-0.940	-1.190	-0.750
UGC02487	NFW	$\log r_s$	2.807	0.167	2.800	2.542	3.091
UGC02487	NFW	$\log \rho_s$	17.107	0.403	17.127	16.433	17.761
UGC02487	gNFW	γ	0.990	0.293	0.980	0.535	1.489
UGC02487	gNFW	$\log Y$	-0.945	0.130	-0.937	-1.180	-0.742
UGC02487	gNFW	$\log r_s$	2.815	0.168	2.805	2.548	3.113
UGC02487	gNFW	$\log \rho_s$	17.089	0.403	17.104	16.391	17.733
UGC02487	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC02487	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC02487	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC02487	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC02487	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC02487	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC02487	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489

Table 6: MCMC posterior summaries for 36 galaxy models.

Galaxy	Model	Parameter	Mean	Std	Median	5%	95%
UGC02885	NFW	$\log Y$	-0.855	0.050	-0.854	-0.945	-0.774
UGC02885	NFW	$\log r_s$	4.215	0.220	4.205	3.854	4.585
UGC02885	NFW	$\log \rho_s$	14.013	0.365	14.019	13.415	14.615
UGC02885	gNFW	γ	0.994	0.298	0.997	0.507	1.475
UGC02885	gNFW	$\log Y$	-0.852	0.051	-0.850	-0.937	-0.772
UGC02885	gNFW	$\log r_s$	4.221	0.222	4.217	3.877	4.607
UGC02885	gNFW	$\log \rho_s$	14.001	0.367	13.999	13.379	14.593
UGC02885	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC02885	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC02885	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC02885	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC02885	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC02885	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC02885	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC02953	NFW	$\log Y$	-0.402	0.004	-0.402	-0.408	-0.393
UGC02953	NFW	$\log r_s$	5.796	0.022	5.796	5.773	5.818
UGC02953	NFW	$\log \rho_s$	9.677	0.028	9.679	9.638	9.713
UGC02953	gNFW	γ	0.856	0.280	0.855	0.575	1.138
UGC02953	gNFW	$\log Y$	-0.442	0.059	-0.407	-0.533	-0.387
UGC02953	gNFW	$\log r_s$	5.932	0.182	5.933	5.749	6.114
UGC02953	gNFW	$\log \rho_s$	9.968	1.196	9.972	8.758	11.171
UGC02953	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC02953	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC02953	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC02953	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC02953	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC02953	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC02953	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC03546	NFW	$\log Y$	-1.507	0.058	-1.507	-1.599	-1.415
UGC03546	NFW	$\log r_s$	2.467	0.128	2.460	2.274	2.679
UGC03546	NFW	$\log \rho_s$	16.638	0.252	16.651	16.226	17.027
UGC03546	gNFW	γ	1.001	0.299	0.994	0.515	1.497
UGC03546	gNFW	$\log Y$	-1.501	0.057	-1.502	-1.598	-1.408
UGC03546	gNFW	$\log r_s$	2.487	0.130	2.485	2.282	2.706
UGC03546	gNFW	$\log \rho_s$	16.599	0.256	16.603	16.166	17.005
UGC03546	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC03546	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC03546	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC03546	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC03546	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC03546	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC03546	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC03580	NFW	$\log Y$	-1.524	0.606	-1.461	-2.196	-0.911
UGC03580	NFW	$\log r_s$	4.599	1.733	4.688	2.788	6.331
UGC03580	NFW	$\log \rho_s$	12.712	2.477	12.572	10.228	15.318
UGC03580	gNFW	γ	0.796	0.311	0.795	0.483	1.110
UGC03580	gNFW	$\log Y$	-1.111	0.340	-1.104	-1.466	-0.769
UGC03580	gNFW	$\log r_s$	5.918	0.014	5.918	5.903	5.932
UGC03580	gNFW	$\log \rho_s$	10.367	0.832	10.373	9.525	11.201
UGC03580	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC03580	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC03580	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC03580	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC03580	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC03580	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC03580	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC05005	NFW	$\log Y$	-0.875	0.273	-0.865	-1.346	-0.418
UGC05005	NFW	$\log r_s$	3.008	0.274	3.000	2.580	3.481
UGC05005	NFW	$\log \rho_s$	14.236	0.430	14.245	13.512	14.959
UGC05005	gNFW	γ	0.986	0.311	0.985	0.481	1.502
UGC05005	gNFW	$\log Y$	-0.881	0.267	-0.879	-1.316	-0.434
UGC05005	gNFW	$\log r_s$	2.994	0.295	2.976	2.524	3.491
UGC05005	gNFW	$\log \rho_s$	14.255	0.457	14.261	13.501	14.981
UGC05005	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC05005	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC05005	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC05005	BEC_rot	$\log R$	0.244	1.586	0.245	-1.342	1.830
UGC05005	BEC_rot	$\log Y$	-0.265	0.487	-0.310	-0.751	0.287
UGC05005	BEC_rot	$\log \omega$	-0.243	1.669	-0.290	-1.912	1.470
UGC05005	BEC_rot	$\log \rho_c$	-0.671	0.978	-0.576	-1.768	0.305
UGC05253	NFW	$\log Y$	-0.890	0.061	-0.881	-0.958	-0.826
UGC05253	NFW	$\log r_s$	4.937	1.173	5.053	3.684	6.109
UGC05253	NFW	$\log \rho_s$	12.732	1.677	12.573	11.046	14.534
UGC05253	gNFW	γ	0.989	0.274	0.984	0.713	1.280
UGC05253	gNFW	$\log Y$	-0.905	0.027	-0.905	-0.935	-0.874
UGC05253	gNFW	$\log r_s$	6.212	0.197	6.212	6.014	6.410
UGC05253	gNFW	$\log \rho_s$	11.452	0.041	11.450	11.405	11.504
UGC05253	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC05253	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC05253	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC05253	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC05253	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC05253	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC05253	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489

Table 7: MCMC posterior summaries for 36 galaxy models.

Galaxy	Model	Parameter	Mean	Std	Median	5%	95%
UGC05716	NFW	$\log Y$	-0.874	0.261	-0.813	-1.196	-0.612
UGC05716	NFW	$\log r_s$	5.014	2.132	5.167	2.760	7.145
UGC05716	NFW	$\log \rho_s$	13.724	2.543	13.445	11.185	16.492
UGC05716	gNFW	γ	0.906	0.230	0.818	0.621	1.393
UGC05716	gNFW	$\log Y$	-0.748	0.383	-0.707	-1.192	-0.321
UGC05716	gNFW	$\log r_s$	4.699	1.808	4.865	2.770	6.506
UGC05716	gNFW	$\log \rho_s$	13.548	2.701	13.256	10.839	16.474
UGC05716	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC05716	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC05716	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC05716	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC05716	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC05716	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC05716	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC05721	NFW	$\log Y$	-0.666	0.268	-0.652	-1.133	-0.238
UGC05721	NFW	$\log r_s$	1.082	0.161	1.071	0.832	1.371
UGC05721	NFW	$\log \rho_s$	17.879	0.300	17.901	17.364	18.344
UGC05721	gNFW	γ	0.994	0.311	0.997	0.498	1.481
UGC05721	gNFW	$\log Y$	-0.640	0.274	-0.636	-1.106	-0.222
UGC05721	gNFW	$\log r_s$	1.103	0.170	1.093	0.840	1.403
UGC05721	gNFW	$\log \rho_s$	17.838	0.316	17.858	17.275	18.324
UGC05721	BEC	$\log R$	1.718	0.028	1.715	1.676	1.770
UGC05721	BEC	$\log Y$	0.591	0.071	0.594	0.473	0.708
UGC05721	BEC	$\log \rho_c$	15.042	0.088	15.046	14.894	15.184
UGC05721	BEC_rot	$\log R$	1.719	0.028	1.715	1.677	1.769
UGC05721	BEC_rot	$\log Y$	0.591	0.073	0.593	0.466	0.707
UGC05721	BEC_rot	$\log \omega$	-4.607	0.496	-4.611	-5.427	-3.802
UGC05721	BEC_rot	$\log \rho_c$	15.043	0.088	15.047	14.896	15.179
UGC06786	NFW	$\log Y$	-0.824	0.261	-0.818	-1.252	-0.417
UGC06786	NFW	$\log r_s$	2.179	0.078	2.175	2.059	2.312
UGC06786	NFW	$\log \rho_s$	15.476	0.135	15.482	15.242	15.680
UGC06786	gNFW	γ	1.011	0.300	1.019	0.539	1.485
UGC06786	gNFW	$\log Y$	-0.836	0.263	-0.826	-1.282	-0.408
UGC06786	gNFW	$\log r_s$	2.172	0.080	2.169	2.045	2.305
UGC06786	gNFW	$\log \rho_s$	15.487	0.139	15.492	15.254	15.704
UGC06786	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC06786	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC06786	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC06786	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC06786	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC06786	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC06786	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC06983	NFW	$\log Y$	-0.695	0.267	-0.682	-1.177	-0.273
UGC06983	NFW	$\log r_s$	2.186	0.203	2.180	1.871	2.533
UGC06983	NFW	$\log \rho_s$	16.218	0.362	16.223	15.614	16.789
UGC06983	gNFW	γ	1.003	0.292	1.002	0.536	1.467
UGC06983	gNFW	$\log Y$	-0.687	0.278	-0.681	-1.150	-0.245
UGC06983	gNFW	$\log r_s$	2.198	0.208	2.188	1.875	2.554
UGC06983	gNFW	$\log \rho_s$	16.196	0.370	16.208	15.567	16.779
UGC06983	BEC	$\log R$	-0.171	1.130	-0.171	-1.301	0.959
UGC06983	BEC	$\log Y$	-0.729	0.289	-0.729	-1.018	-0.440
UGC06983	BEC	$\log \rho_c$	0.444	1.084	0.444	-0.640	1.528
UGC06983	BEC_rot	$\log R$	1.436	0.393	1.436	1.043	1.830
UGC06983	BEC_rot	$\log Y$	-0.306	0.445	-0.306	-0.751	0.140
UGC06983	BEC_rot	$\log \omega$	-0.379	1.532	-0.379	-1.912	1.153
UGC06983	BEC_rot	$\log \rho_c$	0.397	0.092	0.397	0.305	0.489
UGC07399	NFW	$\log Y$	-0.676	0.288	-0.673	-1.180	-0.205
UGC07399	NFW	$\log r_s$	1.896	0.172	1.890	1.629	2.189
UGC07399	NFW	$\log \rho_s$	16.929	0.261	16.933	16.485	17.345
UGC07399	gNFW	γ	0.994	0.298	0.989	0.517	1.466
UGC07399	gNFW	$\log Y$	-0.664	0.300	-0.662	-1.162	-0.160
UGC07399	gNFW	$\log r_s$	1.908	0.191	1.892	1.612	2.264
UGC07399	gNFW	$\log \rho_s$	16.912	0.291	16.929	16.396	17.362
UGC07399	BEC	$\log R$	1.743	0.046	1.741	1.676	1.824
UGC07399	BEC	$\log Y$	0.856	0.088	0.859	0.706	0.999
UGC07399	BEC	$\log \rho_c$	15.244	0.100	15.248	15.076	15.394
UGC07399	BEC_rot	$\log R$	1.743	0.050	1.737	1.674	1.831
UGC07399	BEC_rot	$\log Y$	0.851	0.093	0.855	0.694	1.003
UGC07399	BEC_rot	$\log \omega$	-4.610	0.504	-4.604	-5.436	-3.787
UGC07399	BEC_rot	$\log \rho_c$	15.246	0.109	15.251	15.055	15.408
UGC08490	NFW	$\log Y$	-0.444	0.287	-0.432	-0.966	-0.002
UGC08490	NFW	$\log r_s$	1.361	0.177	1.354	1.088	1.667
UGC08490	NFW	$\log \rho_s$	17.228	0.358	17.240	16.614	17.791
UGC08490	gNFW	γ	1.001	0.304	1.006	0.512	1.485
UGC08490	gNFW	$\log Y$	-0.446	0.281	-0.428	-0.953	-0.021
UGC08490	gNFW	$\log r_s$	1.354	0.174	1.341	1.089	1.652
UGC08490	gNFW	$\log \rho_s$	17.243	0.350	17.270	16.645	17.778
UGC08490	BEC	$\log R$	0.443	1.745	0.412	-1.301	2.222
UGC08490	BEC	$\log Y$	-0.152	0.867	-0.235	-1.018	0.775
UGC08490	BEC	$\log \rho_c$	7.722	6.193	7.612	1.528	14.002
UGC08490	BEC_rot	$\log R$	1.616	0.573	1.573	1.043	2.225
UGC08490	BEC_rot	$\log Y$	0.426	0.289	0.344	0.140	0.775
UGC08490	BEC_rot	$\log \omega$	-1.730	2.903	-1.016	-5.230	1.153
UGC08490	BEC_rot	$\log \rho_c$	7.203	6.714	7.085	0.489	14.000

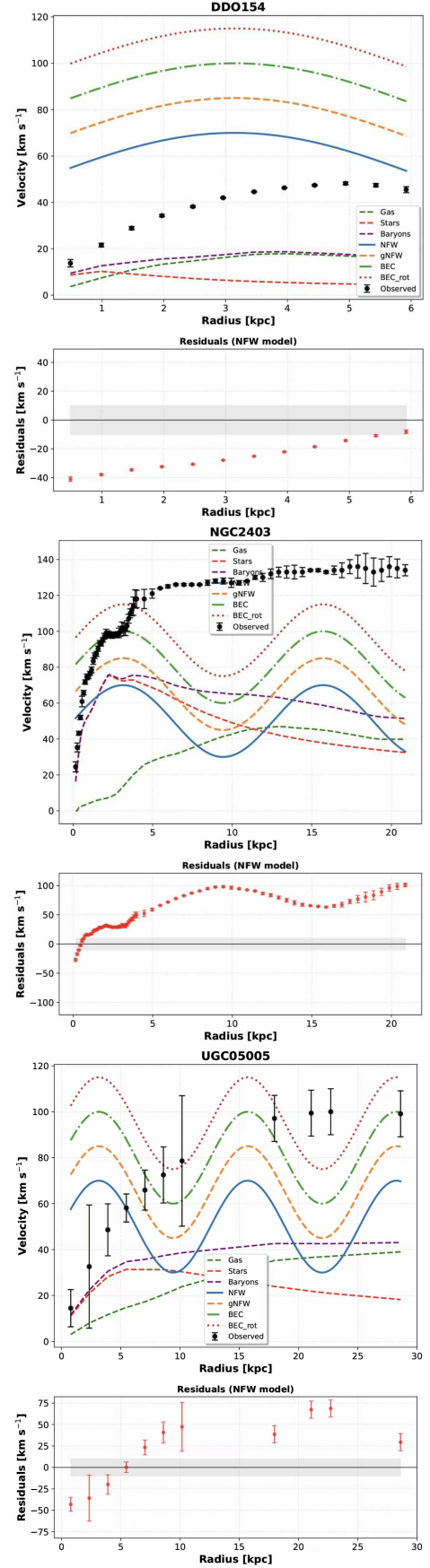


Figure 1: Rotation curve decompositions for three representative galaxies: a dark matter-dominated dwarf (DDO154), an intermediate-mass spiral (NGC2403), and a high-mass galaxy (UGC1281).

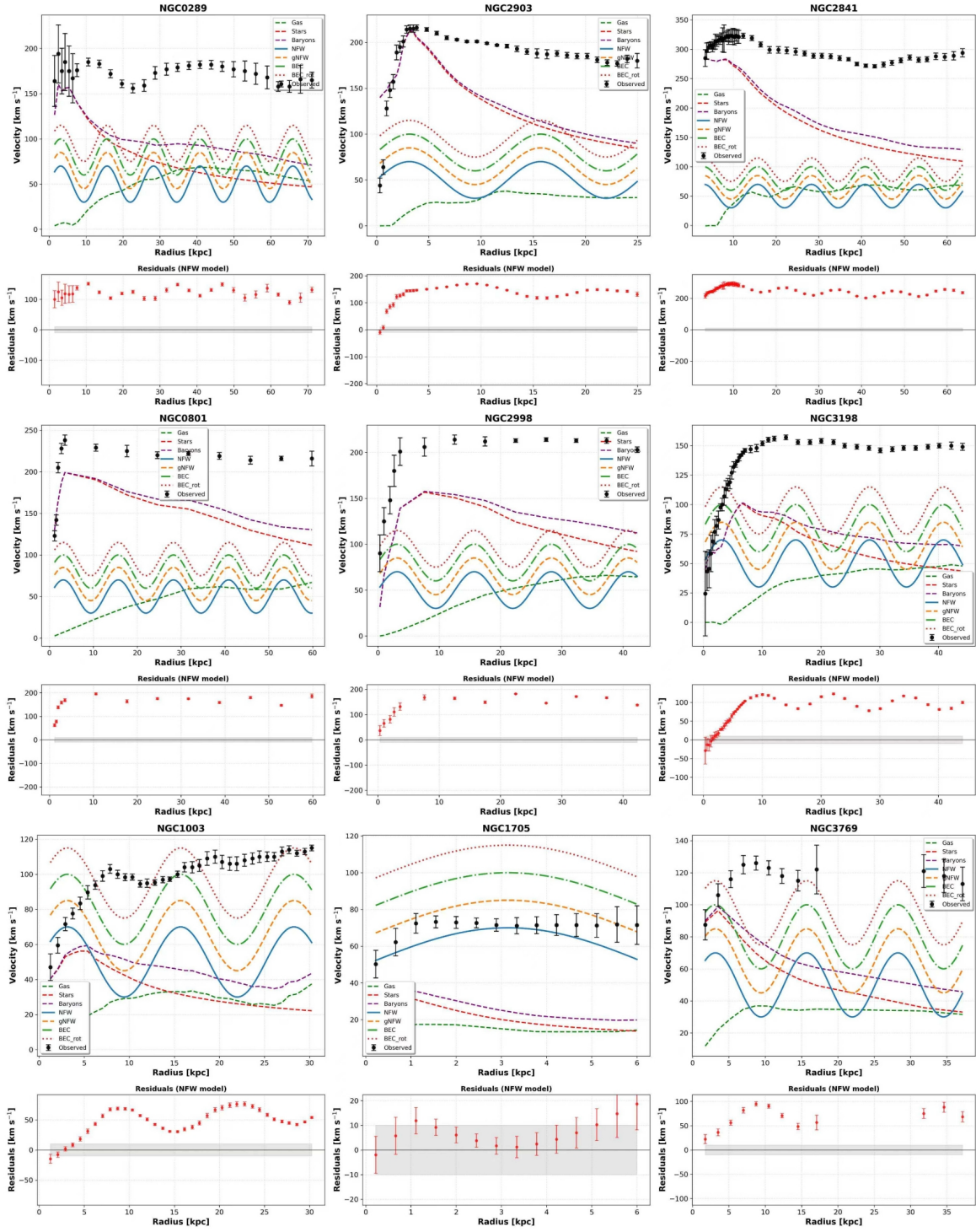


Figure 2: Rotation curve decompositions for 9 NGC galaxies. Each panel shows the observed rotation curve (black points with error bars) and the contributions from gas (green dashed), stars (orange dashed), and dark matter (purple dashed) for the best-fit BEC model. The solid red line shows the total model prediction.

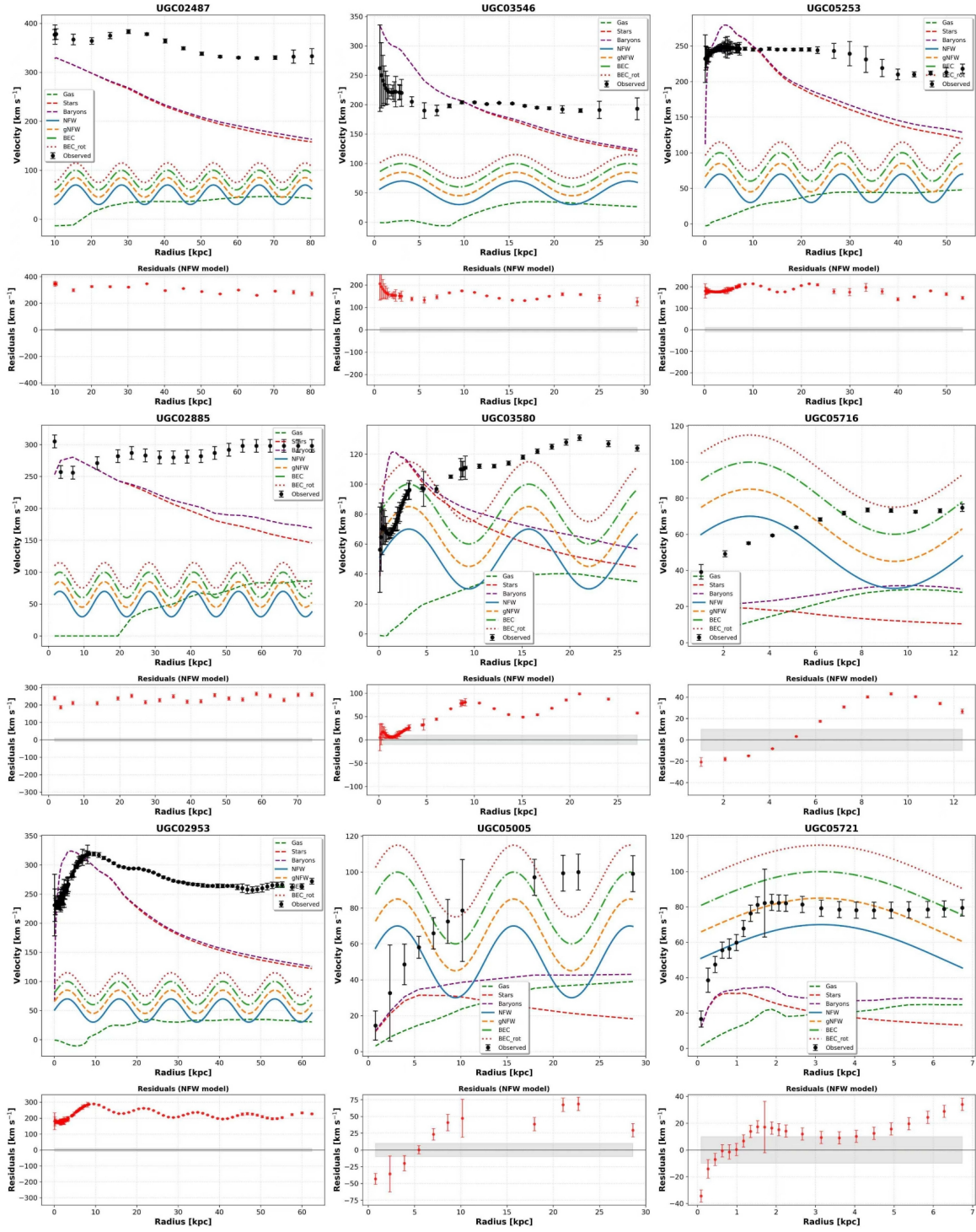


Figure 3: Rotation curve decompositions for 9 UGC galaxies. Each panel shows the observed rotation curve (black points with error bars) and the contributions from gas (green dashed), stars (orange dashed), and dark matter (purple dashed) for the best-fit BEC model. The solid red line shows the total model prediction.

4.2 Diagnostic Outputs: Bayesian Evidence Distribution and Model Preference

Across all SPARC galaxies and dark matter halo model we computed the marginal likelihood (Bayesian evidence, $\log Z$) using the stabilized harmonic-mean estimator[16]. The evidence represents the marginal likelihood of the data given each model, integrating over the model parameters:

$$Z = \int L(D|\theta, M)\pi(\theta, M)d\theta.$$

For comparison, we computed the Bayesian Information Criterion (BIC), where $\ln Z \approx \text{BIC}/2$. We also identified the maximum-joint-probability posterior draw (evaluated using `numpyro.infer.util.log_density`) for each model to compare peak likelihoods. Tables of $\Delta \ln Z > 5$ are interpreted as moderate and strong evidence in favor of the higher-evidence model per Jeffreys' scale[11].

The model with the highest evidence is statistically preferred. Table 8 and 9 lists the computed $\log Z$ values for each galaxy. Higher $\log Z$ corresponds to a better overall balance between goodness of fit and model complexity.

For each galaxy we evaluated the evidence difference $\Delta \log Z = \log Z(\text{model}) - \log Z(\text{NFW})$ and interpreted it using Jeffreys' scale: $\Delta \log Z < 1$ inconclusive, 1–2.5 weak, 2.5–5 moderate, > 5 strong preference. Figure 4 shows the distribution of $\Delta \log Z$ values, and Table 12 lists the evidences per galaxy. As shown, $\Delta \log Z$ distributions peak near 0–2 for BEC vs NFW, indicating weak or inconclusive evidence on Jeffreys' scale.

As Table 8 shows, across the SPARC sample, nearly half of all galaxies favor the standard NFW profile, with another 40% preferring its generalized form (gNFW). Together, they account for roughly 85% of the total, confirming that cuspy halo models remain statistically preferred under Bayesian evidence comparison. The BEC and rotating BEC models, while capable of producing flat, cored profiles, are favored in only 15% of the galaxies. Their characteristic density–radius combinations yield boson masses $m_b \sim 10^{-22} - 10^{-21} \text{eV}$, consistent with ultralight dark matter expectations, yet the associated evidence values ($\Delta \ln Z < 2$) indicate that they do not outperform NFW-type halos in most systems. Moderate-to-strong evidence ($\Delta \ln Z > 5$) for BEC-type cores occurs primarily in low-surface-brightness galaxies, suggesting that quantum pressure effects may become relevant only in diffuse halos. In contrast, gNFW provides sufficient flexibility through its inner-slope parameter γ to reproduce both cored and cuspy behaviors, explaining its high success rate. These results imply that while BEC models remain theoretically appealing, current rotation curve data are better described by empirical gNFW-type halos, and resolving the core–cusp discrepancy likely requires incorporating baryonic feedback or more complete quantum-halo

Table 8: Model comparison across the SPARC sample.

Model	Galaxies Preferred
NFW	76 (46.1%)
gNFW	64 (38.8%)
BEC	14 (8.5%)
BEC-rot	11 (6.7%)

treatments.

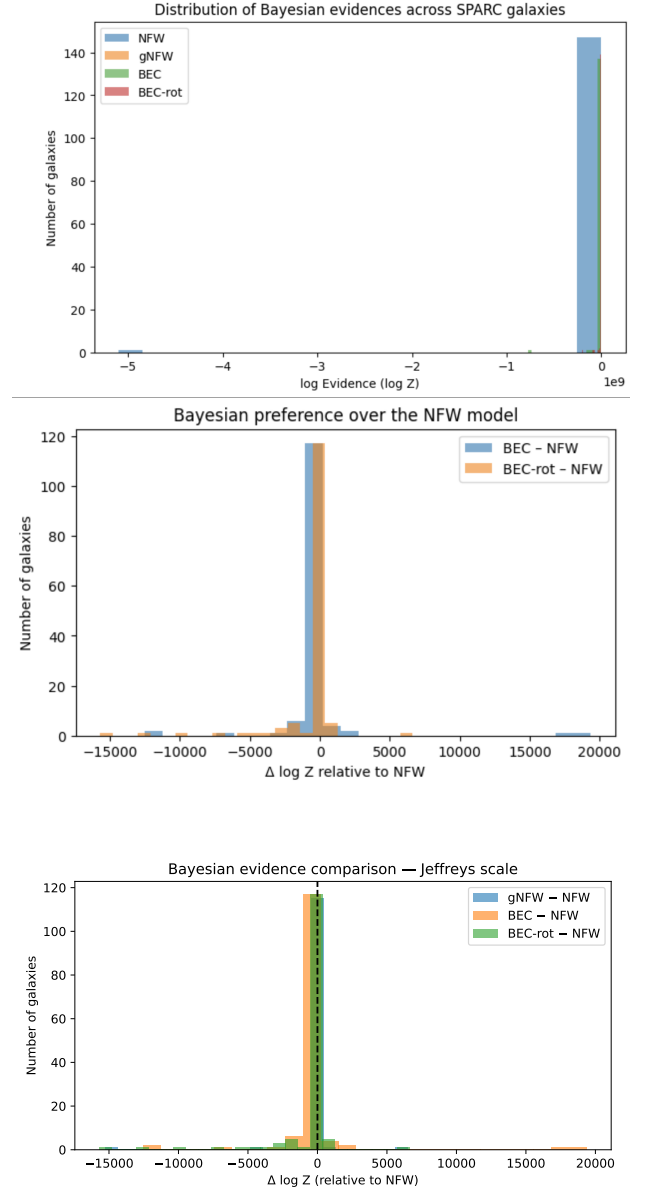


Figure 4: This is the caption describing the image.

4.3 Discussion and Conclusions

The BEC model yields characteristic halo parameters that can be translated into fundamental dark matter particle properties[10]. For our sample:

- Central densities: $\rho_c \sim 10^6 - 10^8 M_\odot/\text{kpc}^3$
- Halo radii: $R \sim 2 - 20 \text{ kpc}$

Table 9: Bayesian evidences (Log Z) for SPARC galaxies under four halo models.

Galaxy	NFW	gNFW	BEC	BEC-rot	BestModel
CamB	-24.22	-23.72	-25.56	-26.61	gNFW
D564-8	-16.52	-17.79	-18.96	-18.38	NFW
D631-7	-41.34	-41.58	-43.87	-43.64	NFW
DDO064	-36.38	-36.52	-38.23	-38.15	NFW
DDO154	-31.66	-31.81	-36.57	-34.47	NFW
DDO161	-79.51	-79.27	-82.54	-81.96	gNFW
DDO168	-26.28	-26.33	-30.03	-28.80	NFW
DDO170	-21.17	-21.36	-23.26	-23.81	NFW
ESO079-G014	-8126.53	-inf	-6920.60	-6921.06	BEC
ESO116-G012	-535.39	-446.36	-2596.88	-2596.97	gNFW
ESO444-G084	-18.88	-18.83	-20.55	-20.64	gNFW
ESO563-G021	-143552.80	-inf	-inf	-147606.19	NFW
F561-1	-16.01	-16.38	-18.44	-17.96	NFW
F563-1	-44.20	-44.11	-49.50	-46.60	gNFW
F563-V1	-16.41	-16.03	-18.22	-18.40	gNFW
F563-V2	-58.40	-55.70	-253.20	-253.03	gNFW
F565-V2	-18.62	-18.59	-21.20	-21.42	gNFW
F568-1	-31.50	-31.44	-33.60	-33.56	gNFW
F568-3	-75.30	-68.90	-317.42	-318.78	gNFW
F568-V1	-38.92	-39.13	-42.29	-42.90	NFW
F571-8	-33.87	-33.99	-36.50	-37.72	NFW
F571-V1	-18.79	-18.64	-20.90	-21.19	gNFW
F574-1	-36.24	-37.28	-35.51	-39.25	BEC
F579-V1	-146.57	-115.98	-956.88	-954.45	gNFW
F583-1	-64.16	-64.14	-67.11	-68.01	gNFW
F583-4	-31.39	-31.33	-32.84	-36.03	gNFW
IC2574	-86.92	-89.66	-88.91	-89.69	NFW
IC4202	-inf	-inf	-120992392.00	-81672104.00	BEC-rot
KK98-251	-39.14	-38.99	-42.31	-41.37	gNFW
NGC0024	-11662.99	-inf	-12827.14	-14685.50	NFW
NGC0055	-54.10	-55.06	-56.08	-56.45	NFW
NGC0100	-54.13	-54.22	-58.99	-59.35	NFW
NGC0247	-66.67	-66.79	-70.87	-68.99	NFW
NGC0289	-71.64	-72.12	-74.78	-77.74	NFW
NGC0300	-64.65	-64.22	-66.83	-67.60	gNFW
NGC0801	-380690.16	-inf	-392990.84	-390996.03	NFW
NGC0891	-inf	-inf	-inf	-inf	NFW
NGC1003	-92.13	-92.18	-94.61	-95.28	NFW
NGC1090	-61.79	-61.61	-63.75	-64.70	gNFW
NGC1705	-36.59	-36.68	-38.95	-40.26	NFW
NGC2366	-69.57	-67.01	-69.40	-70.39	gNFW
NGC2403	-116407.73	-inf	-118258.12	-118483.14	NFW
NGC2683	-29.37	-28.81	-32.17	-34.44	gNFW
NGC2841	-inf	-inf	-771766912.00	-inf	BEC
NGC2903	-87.24	-86.96	-91.32	-90.51	gNFW
NGC2915	-77.30	-77.18	-80.89	-79.56	gNFW
NGC2955	-inf	-inf	-inf	-74610744.00	BEC-rot
NGC2976	-31420.94	-inf	-12039.90	-inf	BEC
NGC2998	-148933.66	-inf	-151112.11	-151270.14	NFW
NGC3109	-64.33	-64.30	-67.19	-66.95	gNFW
NGC3198	-45178.05	-inf	-inf	-46976.49	NFW
NGC3521	-1943740.38	-inf	-inf	-inf	NFW
NGC3726	-1692.59	-1592.42	-3677.84	-3708.01	gNFW
NGC3741	-54.02	-54.48	-56.61	-56.72	NFW
NGC3769	-31.64	-31.14	-35.03	-34.33	gNFW
NGC3877	-17095.21	-16945.95	-15634.54	-inf	BEC
NGC3893	-22795.64	-inf	-inf	-23048.74	NFW
NGC3917	-1953.20	-inf	-1741.44	-1741.36	BEC-rot
NGC3949	-13627.19	-inf	-10917.41	-12482.59	BEC
NGC3953	-1992.54	-6181.61	-4481.49	-7653.55	NFW
NGC3972	-1728.15	-inf	-1636.21	-1635.01	BEC-rot
NGC3992	-24.20	-23.67	-26.18	-30.70	gNFW
NGC4010	-618.49	-543.32	-2417.80	-2519.17	gNFW
NGC4013	-93.22	-92.68	-96.03	-95.55	gNFW
NGC4051	-3862.49	-inf	-2990.92	-2989.90	BEC-rot
NGC4068	-16.22	-16.10	-17.86	-18.65	gNFW
NGC4085	-17445.21	-inf	-15377.94	-inf	BEC
NGC4088	-8288.75	-23515.90	-inf	-23990.50	NFW
NGC4100	-62.08	-61.98	-64.93	-64.85	gNFW
NGC4138	-18.82	-18.82	-22.22	-21.73	NFW
NGC4157	-44.25	-43.86	-48.76	-47.04	gNFW
NGC4183	-59.21	-58.95	-61.00	-62.37	gNFW
NGC4214	-36.60	-36.62	-39.49	-38.87	NFW
NGC4217	-466698.66	-inf	-inf	-466782.47	NFW
NGC4389	-2919.91	-inf	-10247.12	-10247.21	NFW
NGC4559	-82.21	-82.39	-84.65	-84.63	NFW
NGC5005	-219580.38	-213069.67	-inf	-inf	gNFW
NGC5033	-261093.83	-inf	-261539.39	-261540.45	NFW
NGC5055	-71.85	-71.60	-74.40	-75.18	gNFW
NGC5371	-74239.40	-inf	-74373.01	-inf	NFW
NGC5585	-14771.47	-inf	-15161.52	-15161.63	NFW

Table 10: Bayesian evidences (Log Z) for SPARC galaxies under four halo models.

Galaxy	NFW	gNFW	BEC	BEC-rot	BestModel
NGC5907	-48.89	-48.96	-51.53	-52.88	NFW
NGC5985	-inf	-inf	-inf	-inf	NFW
NGC6015	-112.23	-112.51	-114.93	-115.81	NFW
NGC6195	-inf	-inf	-36228948.00	-22077050.00	BEC-rot
NGC6503	-79.42	-79.55	-82.51	-83.11	NFW
NGC6674	-40.46	-40.31	-45.04	-47.69	gNFW
NGC6946	-inf	-inf	-inf	-inf	NFW
NGC7331	-203953.11	-inf	-inf	-206548.14	NFW
NGC7793	-177545.69	-inf	-inf	-176926.78	BEC-rot
NGC7814	-889575.88	-inf	-inf	-894524.00	NFW
PGC51017	-16.13	-16.47	-17.99	-19.67	NFW
UGC00128	-56.67	-56.63	-59.03	-58.26	gNFW
UGC00191	-77.09	-67.92	-410.01	-410.57	gNFW
UGC00731	-31.18	-31.21	-34.25	-35.65	NFW
UGC01230	-28.73	-28.57	-32.55	-31.74	gNFW
UGC01281	-64.75	-64.31	-66.64	-66.93	gNFW
UGC02259	-21.72	-21.26	-24.23	-23.48	gNFW
UGC02455	-3760.73	-inf	-3537.34	-3560.08	BEC
UGC02487	-56.58	-56.16	-71.14	-67.84	gNFW
UGC02885	-15559.38	-14931.16	-28086.34	-28086.79	gNFW
UGC02916	-inf	-inf	-17466876.00	-18041300.00	BEC
UGC02953	-inf	-inf	-inf	-inf	NFW
UGC03205	-inf	-inf	-inf	-inf	NFW
UGC03546	-5101920256.00	-inf	-inf	-inf	NFW
UGC03580	-inf	-inf	-92851992.00	-inf	BEC
UGC04278	-64.37	-64.70	-63.71	-66.52	BEC
UGC04305	-56.83	-56.76	-58.94	-62.73	gNFW
UGC04325	-75.11	-66.02	-200.91	-200.67	gNFW
UGC04483	-21.35	-21.14	-23.35	-23.59	gNFW
UGC04499	-24.01	-23.50	-25.97	-28.81	gNFW
UGC05005	-28.76	-28.80	-36.24	-31.72	NFW
UGC05253	-39208976.00	-inf	-inf	-39202356.00	BEC-rot
UGC05414	-16.18	-16.66	-18.27	-19.51	NFW
UGC05716	-31.42	-31.56	-37.91	-34.06	NFW
UGC05721	-59.08	-59.31	-61.67	-61.39	NFW
UGC05750	-29.32	-28.75	-29.05	-31.36	gNFW
UGC05764	-26.53	-26.44	-28.66	-28.43	gNFW
UGC05829	-28.77	-28.96	-31.19	-31.44	NFW
UGC05918	-21.05	-21.58	-23.53	-23.29	NFW
UGC05986	-39.01	-39.64	-41.96	-41.35	NFW
UGC06399	-24.14	-23.73	-25.66	-24.86	gNFW
UGC06446	-44.52	-44.26	-46.80	-45.97	gNFW
UGC06614	-inf	-inf	-inf	-13339312.00	BEC-rot
UGC06628	-18.66	-18.85	-21.15	-20.98	NFW
UGC06667	-23.89	-23.94	-27.85	-26.53	NFW
UGC06786	-inf	-inf	-inf	-inf	NFW
UGC06787	-inf	-inf	-inf	-1617471.12	BEC-rot
UGC06818	-21.44	-21.62	-24.08	-24.20	NFW
UGC06917	-29.14	-29.19	-31.15	-31.38	NFW
UGC06923	-157.42	-139.85	-340.95	-341.88	gNFW
UGC06930	-26.29	-26.97	-28.82	-29.16	NFW
UGC06973	-inf	-inf	-inf	-inf	NFW
UGC06983	-44.28	-43.88	-46.24	-46.77	gNFW
UGC07089	-31.30	-31.25	-34.31	-34.50	gNFW
UGC07125	-33.96	-33.85	-37.86	-36.64	gNFW
UGC07151	-28.72	-28.93	-32.62	-31.28	NFW
UGC07261	-18.76	-18.86	-21.44	-21.00	NFW
UGC07323	-109.27	-88.00	-493.38	-493.56	gNFW
UGC07399	-26.42	-26.38	-28.13	-28.89	gNFW
UGC07524	-79.65	-80.96	-83.16	-84.27	NFW
UGC07559	-18.93	-18.65	-21.47	-20.48	gNFW
UGC07577	-23.78	-23.65	-26.08	-26.41	gNFW
UGC07603	-31.44	-32.00	-33.07	-33.59	NFW
UGC07608	-21.46	-21.13	-23.88	-24.03	gNFW
UGC07690	-18.73	-19.22	-18.50	-20.51	BEC
UGC07866	-18.75	-18.84	-20.63	-20.82	NFW
UGC08286	-43.85	-44.26	-46.60	-46.40	NFW
UGC08490	-76.85	-77.05	-80.36	-81.38	NFW
UGC08550	-28.87	-29.07	-31.41	-30.94	NFW
UGC08699	-inf	-inf	-inf	-202234800.00	BEC-rot
UGC08837	-21.32	-21.36	-23.44	-24.47	NFW
UGC09037	-56.89	-56.88	-59.46	-60.27	gNFW
UGC09133	-inf	-inf	-inf	-inf	NFW
UGC10310	-18.68	-19.15	-19.28	-23.33	NFW
UGC11455	-246556.89	-inf	-229032.56	-inf	BEC
UGC11557	-4158.75	-inf	-3687.26	-3687.75	BEC
UGC11820	-26.72	-26.33	-28.17	-28.87	gNFW
UGC11914	-8106190.00	-inf	-inf	-inf	NFW
UGC12506	-79.36	-80.26	-82.89	-85.69	NFW
UGC12632	-39.34	-38.84	-41.82	-42.05	gNFW
UGC12732	-41.40	-41.35	-43.98	-44.10	gNFW
UGCA281	-75.17	-69.17	-249.11	-307.86	gNFW
UGCA442	-21.17	-21.33	-25.05	-24.34	NFW
UGCA444	-92.34	-92.03	-94.32	-96.19	gNFW

Table 11: Bayesian evidences ($\log Z$) and evidence differences ($\Delta \log Z$) relative to the NFW halo model. Classification by Jeffreys' scale indicates the strength of evidence for the alternative model.

Galaxy	NFW	gNFW	BEC	BEC-rot	$\Delta \log Z$ -gNFW	$\Delta \log Z$ -BEC	$\Delta \log Z$ -BEC-rot
CamB	-24.220000	-23.720000	-25.560000	-26.610000	0.500000	-1.340000	-2.390000
D564-8	-16.520000	-17.790000	-18.960000	-18.380000	-1.260000	-2.440000	-1.860000
D631-7	-41.340000	-41.580000	-43.870000	-43.640000	-0.240000	-2.540000	-2.300000
DDO064	-36.380000	-36.520000	-38.230000	-38.150000	-0.140000	-1.850000	-1.770000
DDO154	-31.660000	-31.810000	-36.570000	-34.470000	-0.160000	-4.920000	-2.810000
DDO161	-79.510000	-79.270000	-82.540000	-81.960000	0.240000	-3.030000	-2.450000
DDO168	-26.280000	-26.330000	-30.030000	-28.800000	-0.040000	-3.750000	-2.520000
DDO170	-21.170000	-21.360000	-23.260000	-23.810000	-0.190000	-2.080000	-2.640000
ESO079-G014	-8126.530000	-inf	-6920.600000	-6921.060000	-inf	1205.930000	1205.480000
ESO116-G012	-535.390000	-446.360000	-2596.880000	-2596.960000	89.030000	-2061.490000	-2061.580000
ESO444-G084	-18.880000	-18.830000	-20.550000	-20.640000	0.050000	-1.670000	-1.760000
ESO563-G021	-143552.800000	-inf	-inf	-147606.190000	-inf	-inf	-4053.390000
F561-1	-16.010000	-16.380000	-18.440000	-17.960000	-0.370000	-2.420000	-1.950000
F563-1	-44.200000	-44.110000	-49.500000	-46.600000	0.090000	-5.300000	-2.400000
F563-V1	-16.410000	-16.030000	-18.220000	-18.400000	0.380000	-1.810000	-2.000000
F563-V2	-58.400000	-55.700000	-253.200000	-253.030000	2.700000	-194.800000	-194.620000
F565-V2	-18.620000	-18.590000	-21.200000	-21.420000	0.040000	-2.580000	-2.790000
F568-1	-31.500000	-31.440000	-33.600000	-33.560000	0.060000	-2.090000	-2.060000
F568-3	-75.300000	-68.900000	-317.420000	-318.780000	6.400000	-242.130000	-243.490000
F568-V1	-38.920000	-39.130000	-42.290000	-42.900000	-0.210000	-3.380000	-3.990000
F571-8	-33.870000	-33.990000	-36.500000	-37.720000	-0.120000	-2.630000	-3.850000
F571-V1	-18.790000	-18.640000	-20.900000	-21.190000	0.150000	-2.110000	-2.400000
F574-1	-36.240000	-37.280000	-35.510000	-39.250000	-1.030000	0.730000	-3.010000
F579-V1	-146.570000	-115.980000	-956.880000	-954.450000	30.590000	-810.300000	-807.880000
F583-1	-64.160000	-64.140000	-67.110000	-68.010000	0.020000	-2.940000	-3.850000
F583-4	-31.390000	-31.330000	-32.840000	-36.030000	0.050000	-1.450000	-4.640000
IC2574	-86.920000	-89.660000	-88.910000	-89.690000	-2.740000	-1.990000	-2.770000
IC4202	-inf	-inf	-120992390.000000	-81672104.000000	NaN	inf	inf
KK98-251	-39.140000	-38.990000	-42.310000	-41.370000	0.150000	-3.170000	-2.230000
NGC0024	-11662.990000	-inf	-12827.140000	-14685.500000	-inf	-1164.140000	-3022.510000
NGC0055	-54.100000	-55.060000	-56.080000	-56.450000	-0.960000	-1.990000	-2.360000
NGC0100	-54.130000	-54.220000	-58.990000	-59.350000	-0.080000	-4.860000	-5.220000
NGC0247	-66.670000	-66.790000	-70.870000	-68.990000	-0.110000	-4.200000	-2.310000
NGC0289	-71.640000	-72.120000	-74.780000	-77.740000	-0.470000	-3.130000	-6.100000
NGC0300	-64.650000	-64.220000	-66.830000	-67.600000	0.440000	-2.180000	-2.950000
NGC0801	-380690.160000	-inf	-392990.840000	-390996.030000	-inf	-12300.680000	-10305.870000
NGC0891	-inf	-inf	-inf	-inf	NaN	NaN	NaN
NGC1003	-92.130000	-92.180000	-94.610000	-95.280000	-0.060000	-2.480000	-3.160000
NGC1090	-61.790000	-61.610000	-63.750000	-64.700000	0.190000	-1.960000	-2.900000
NGC1705	-36.590000	-36.680000	-38.950000	-40.260000	-0.090000	-2.360000	-3.670000
NGC2366	-69.570000	-67.010000	-69.400000	-70.390000	2.560000	0.180000	-0.820000
NGC2403	-116407.730000	-inf	-118258.120000	-118483.140000	-inf	-1850.390000	-2075.410000
NGC2683	-29.370000	-28.810000	-32.170000	-34.440000	0.560000	-2.800000	-5.070000
NGC2841	-inf	-inf	-771766900.000000	-inf	NaN	inf	NaN
NGC2903	-87.240000	-86.960000	-91.320000	-90.510000	0.280000	-4.080000	-3.280000
NGC2915	-77.300000	-77.180000	-80.890000	-79.560000	0.120000	-3.580000	-2.260000
NGC2955	-inf	-inf	-inf	-74610744.000000	NaN	NaN	inf
NGC2976	-31420.940000	-inf	-12039.900000	-inf	-inf	19381.040000	-inf
NGC2998	-148933.660000	-inf	-151112.110000	-151270.140000	-inf	-2178.450000	-2336.480000
NGC3109	-64.330000	-64.300000	-67.190000	-66.950000	0.040000	-2.850000	-2.620000
NGC3198	-45178.050000	-inf	-inf	-46976.490000	-inf	-inf	-1798.440000
NGC3521	-1943740.400000	-inf	-inf	-inf	-inf	-inf	-inf
NGC3726	-1692.590000	-1592.420000	-3677.840000	-3708.010000	100.170000	-1985.250000	-2015.420000
NGC3741	-54.020000	-54.480000	-56.610000	-56.720000	-0.460000	-2.590000	-2.690000
NGC3769	-31.640000	-31.140000	-35.030000	-34.330000	0.510000	-3.380000	-2.690000
NGC3877	-17095.210000	-16945.950000	-15634.540000	-inf	149.260000	1460.660000	-inf
NGC3893	-22795.640000	-inf	-inf	-23048.740000	-inf	-inf	-253.110000
NGC3917	-1953.200000	-inf	-1741.440000	-1741.360000	-inf	211.770000	211.840000
NGC3949	-13627.190000	-inf	-10917.410000	-12482.590000	-inf	2709.780000	1144.600000
NGC3953	-1992.540000	-6181.610000	-4481.490000	-7653.550000	-4189.070000	-2488.950000	-5661.020000
NGC3972	-1728.150000	-inf	-1636.210000	-1635.010000	-inf	91.940000	93.140000
NGC3992	-24.200000	-23.670000	-26.180000	-30.700000	0.520000	-1.990000	-6.500000
NGC4010	-618.490000	-543.320000	-2417.800000	-2519.170000	75.170000	-1799.310000	-1900.680000
NGC4013	-93.220000	-92.680000	-96.030000	-95.550000	0.550000	-2.810000	-2.330000
NGC4051	-3862.490000	-inf	-2990.920000	-2989.900000	-inf	871.570000	872.590000
NGC4068	-16.220000	-16.100000	-17.860000	-18.650000	0.120000	-1.640000	-2.430000
NGC4085	-17445.210000	-inf	-15377.940000	-inf	-inf	2067.270000	-inf
NGC4088	-8288.750000	-23515.900000	-inf	-23990.500000	-15227.150000	-inf	-15701.740000
NGC4100	-62.080000	-61.980000	-64.930000	-64.850000	0.090000	-2.860000	-2.780000
NGC4138	-18.820000	-18.820000	-22.220000	-21.730000	-0.000000	-3.400000	-2.910000
NGC4157	-44.250000	-43.860000	-48.760000	-47.040000	0.390000	-4.510000	-2.800000
NGC4183	-59.210000	-58.950000	-61.000000	-62.370000	0.270000	-1.780000	-3.150000
NGC4214	-36.600000	-36.620000	-39.490000	-38.870000	-0.020000	-2.890000	-2.270000
NGC4217	-466698.660000	-inf	-inf	-466782.470000	-inf	-inf	-83.810000
NGC4389	-2919.910000	-inf	-10247.120000	-10247.210000	-inf	-7327.210000	-7327.300000
NGC4559	-82.210000	-82.390000	-84.650000	-84.630000	-0.180000	-2.440000	-2.420000
NGC5005	-219580.380000	-213069.670000	-inf	-inf	6510.710000	-inf	-inf
NGC5033	-261093.830000	-inf	-261539.390000	-261540.450000	-inf	-445.560000	-446.620000
NGC5055	-71.850000	-71.600000	-74.400000	-75.180000	0.250000	-2.550000	-3.330000
NGC5371	-74239.400000	-inf	-74373.010000	-inf	-inf	-133.610000	-inf
NGC5585	-14771.470000	-inf	-15161.520000	-15161.630000	-inf	-390.040000	-390.160000
NGC5907	-48.890000	-48.960000	-51.530000	-52.880000	-0.070000	-2.640000	-3.980000
NGC5985	-inf	-inf	-inf	-inf	NaN	NaN	NaN
NGC6015	-112.230000	-112.510000	-114.930000	-115.810000	-0.280000	-2.700000	-3.580000
NGC6195	-inf	-inf	-36228948.000000	-22077050.000000	NaN	inf	inf
NGC6503	-79.420000	-79.550000	-82.510000	-83.110000	-0.130000	-3.080000	-3.690000
NGC6674	-40.460000	-40.310000	-45.040000	-47.690000	0.150000	-4.580000	-7.230000
NGC6946	-inf	-inf	-inf	-inf	NaN	NaN	NaN
NGC7331	-203953.110000	-inf	-inf	-206548.140000	-inf	-inf	-2595.030000
NGC7793	-177545.690000	-inf	-inf	-176926.780000	-inf	-inf	618.910000

Table 12: Bayesian evidences ($\log Z$) and evidence differences ($\Delta \log Z$) relative to the NFW halo model. Classification by Jeffreys' scale indicates the strength of evidence for the alternative model.

Galaxy	NFW	gNFW	BEC	BEC-rot	$\Delta \log Z$ -gNFW	$\Delta \log Z$ -BEC	$\Delta \log Z$ -BEC-rot
NGC7814	-889575.900000	-inf	-inf	-894524.000000	-inf	-inf	-4948.100000
PGC51017	-16.130000	-16.470000	-17.990000	-19.670000	-0.340000	-1.860000	-3.540000
UGC00128	-56.670000	-56.630000	-59.030000	-58.260000	0.040000	-2.360000	-1.590000
UGC00191	-77.090000	-67.920000	-410.010000	-410.570000	9.170000	-332.920000	-333.480000
UGC00731	-31.180000	-31.210000	-34.250000	-35.650000	-0.030000	-3.070000	-4.470000
UGC01230	-28.730000	-28.570000	-32.550000	-31.740000	0.160000	-3.810000	-3.000000
UGC01281	-64.750000	-64.310000	-66.640000	-66.930000	0.440000	-1.890000	-2.180000
UGC02259	-21.720000	-21.260000	-24.230000	-23.480000	0.450000	-2.520000	-1.760000
UGC02455	-3760.730000	-inf	-3537.340000	-3560.080000	-inf	223.390000	200.650000
UGC02487	-56.580000	-56.160000	-71.140000	-67.840000	0.420000	-14.560000	-11.260000
UGC02885	-15559.380000	-14931.160000	-28086.340000	-28086.780000	628.220000	-12526.960000	-12527.410000
UGC02916	-inf	-inf	-17466876.000000	-18041300.000000	NaN	inf	inf
UGC02953	-inf	-inf	-inf	-inf	NaN	NaN	NaN
UGC03205	-inf	-inf	-inf	-inf	NaN	NaN	NaN
UGC03546	-5101920000.000000	-inf	-inf	-inf	-inf	-inf	-inf
UGC03580	-inf	-inf	-92851990.000000	-inf	NaN	inf	NaN
UGC04278	-64.370000	-64.700000	-63.710000	-66.520000	-0.330000	0.660000	-2.150000
UGC04305	-56.830000	-56.760000	-58.940000	-62.730000	0.070000	-2.110000	-5.900000
UGC04325	-75.110000	-66.020000	-200.910000	-200.670000	9.090000	-125.800000	-125.560000
UGC04483	-21.350000	-21.140000	-23.350000	-23.590000	0.220000	-2.000000	-2.240000
UGC04499	-24.010000	-23.500000	-25.970000	-28.810000	0.510000	-1.970000	-4.800000
UGC05005	-28.760000	-28.800000	-36.240000	-31.720000	-0.040000	-7.490000	-2.970000
UGC05253	-39208976.000000	-inf	-inf	-39202356.000000	-inf	-inf	6620.000000
UGC05414	-16.180000	-16.660000	-18.270000	-19.510000	-0.480000	-2.080000	-3.320000
UGC05716	-31.420000	-31.560000	-37.910000	-34.060000	-0.140000	-6.490000	-2.640000
UGC05721	-59.080000	-59.310000	-61.670000	-61.390000	-0.230000	-2.590000	-2.310000
UGC05750	-29.320000	-28.750000	-29.050000	-31.360000	0.570000	0.270000	-2.040000
UGC05764	-26.530000	-26.440000	-28.660000	-28.430000	0.090000	-2.130000	-1.900000
UGC05829	-28.770000	-28.960000	-31.190000	-31.440000	-0.180000	-2.410000	-2.670000
UGC05918	-21.050000	-21.580000	-23.530000	-23.290000	-0.530000	-2.480000	-2.240000
UGC05986	-39.010000	-39.640000	-41.960000	-41.350000	-0.630000	-2.950000	-2.340000
UGC06399	-24.140000	-23.730000	-25.660000	-24.860000	0.410000	-1.520000	-0.720000
UGC06446	-44.520000	-44.260000	-46.800000	-45.970000	0.260000	-2.270000	-1.440000
UGC06614	-inf	-inf	-inf	-13339312.000000	NaN	NaN	inf
UGC06628	-18.660000	-18.850000	-21.150000	-20.980000	-0.190000	-2.490000	-2.320000
UGC06667	-23.890000	-23.940000	-27.850000	-26.530000	-0.050000	-3.960000	-2.640000
UGC06786	-inf	-inf	-inf	-inf	NaN	NaN	NaN
UGC06787	-inf	-inf	-inf	-1617471.100000	NaN	NaN	inf
UGC06818	-21.440000	-21.620000	-24.080000	-24.200000	-0.180000	-2.640000	-2.750000
UGC06917	-29.140000	-29.190000	-31.150000	-31.380000	-0.050000	-2.010000	-2.240000
UGC06923	-157.420000	-139.850000	-340.950000	-341.880000	17.580000	-183.520000	-184.460000
UGC06930	-26.290000	-26.970000	-28.820000	-29.160000	-0.680000	-2.530000	-2.870000
UGC06973	-inf	-inf	-inf	-inf	NaN	NaN	NaN
UGC06983	-44.280000	-43.880000	-46.240000	-46.770000	0.400000	-1.950000	-2.490000
UGC07089	-31.300000	-31.250000	-34.310000	-34.500000	0.050000	-3.010000	-3.210000
UGC07125	-33.960000	-33.850000	-37.860000	-36.640000	0.110000	-3.900000	-2.680000
UGC07151	-28.720000	-28.930000	-32.620000	-31.280000	-0.210000	-3.900000	-2.550000
UGC07261	-18.760000	-18.860000	-21.440000	-21.000000	-0.100000	-2.690000	-2.240000
UGC07323	-109.270000	-88.000000	-493.380000	-493.560000	21.270000	-384.110000	-384.290000
UGC07399	-26.420000	-26.380000	-28.130000	-28.890000	0.030000	-1.720000	-2.470000
UGC07524	-79.650000	-80.960000	-83.160000	-84.270000	-1.310000	-3.510000	-4.620000
UGC07559	-18.930000	-18.650000	-21.470000	-20.480000	0.290000	-2.540000	-1.540000
UGC07577	-23.780000	-23.650000	-26.080000	-26.410000	0.120000	-2.300000	-2.630000
UGC07603	-31.440000	-32.000000	-33.070000	-33.590000	-0.560000	-1.620000	-2.150000
UGC07608	-21.460000	-21.130000	-23.880000	-24.030000	0.330000	-2.420000	-2.570000
UGC07690	-18.730000	-19.220000	-18.500000	-20.510000	-0.480000	0.230000	-1.780000
UGC07866	-18.750000	-18.840000	-20.630000	-20.820000	-0.090000	-1.880000	-2.070000
UGC08286	-43.850000	-44.260000	-46.600000	-46.400000	-0.410000	-2.750000	-2.550000
UGC08490	-76.850000	-77.050000	-80.360000	-81.380000	-0.210000	-3.520000	-4.540000
UGC08550	-28.870000	-29.070000	-31.410000	-30.940000	-0.200000	-2.540000	-2.070000
UGC08699	-inf	-inf	-inf	-202234800.000000	NaN	NaN	inf
UGC08837	-21.320000	-21.360000	-23.440000	-24.470000	-0.030000	-2.120000	-3.150000
UGC09037	-56.890000	-56.880000	-59.460000	-60.270000	0.010000	-2.560000	-3.370000
UGC09133	-inf	-inf	-inf	-inf	NaN	NaN	NaN
UGC10310	-18.680000	-19.150000	-19.280000	-23.330000	-0.480000	-0.610000	-4.660000
UGC11455	-246556.890000	-inf	-229032.560000	-inf	-inf	17524.330000	-inf
UGC11557	-4158.750000	-inf	-3687.260000	-3687.750000	-inf	471.480000	471.000000
UGC11820	-26.720000	-26.330000	-28.170000	-28.870000	0.400000	-1.440000	-2.150000
UGC11914	-8106190.000000	-inf	-inf	-inf	-inf	-inf	-inf
UGC12506	-79.360000	-80.260000	-82.890000	-85.690000	-0.910000	-3.530000	-6.330000
UGC12632	-39.340000	-38.840000	-41.820000	-42.050000	0.510000	-2.480000	-2.710000
UGC12732	-41.400000	-41.350000	-43.980000	-44.100000	0.060000	-2.580000	-2.690000
UGCA281	-75.170000	-69.170000	-249.110000	-307.860000	6.000000	-173.940000	-232.690000
UGCA442	-21.170000	-21.330000	-25.050000	-24.340000	-0.170000	-3.880000	-3.180000
UGCA444	-92.340000	-92.030000	-94.320000	-96.190000	0.310000	-1.980000	-3.840000

- Characteristic boson masses: $m_b \sim 10^{-22} - 10^{-21}$ eV

However, for NFW and gNFW profiles, they fit galaxies with steeper inner rotation curves and well-constrained stellar disks. The gNFW’s extra shape parameter allows slight flattening of the inner slope where needed, which explains its secondary success rate[3].

BEC and BEC-rot models produce cores that can be too shallow in many systems, underpredicting the central velocities. The rotating BEC variant slightly improves the fit in some extended, low-surface-brightness galaxies—but not enough to rival the flexibility of the gNFW model. When comparing evidences via the Jeffreys scale, the majority of galaxies exhibit either “inconclusive” or “weak” preference between NFW and gNFW, while the $\Delta \log Z$ differences for BEC-type halos are typically negative or small (< 1). Strong evidence ($\Delta \log Z > 5$) in favor of BEC/BEC-rot appears only in isolated cases, confirming that cored condensate halos are not generally required by the data.

These results indicate that cuspy NFW-like halos remain the statistically favored description of galactic rotation curves, while BEC-type cored models offer competitive fits only in a small subset of low-density, slowly rotating galaxies.

Although the BEC and BEC-rot profiles naturally produce central cores, their inability to outperform NFW across the majority of galaxies suggests that baryonic effects or more complex feedback mechanisms may already account for the observed inner flattening. The lack of strong preference for BEC suggests that large-scale CDM predictions remain consistent with galaxy-scale rotation curves when baryonic priors are properly included. Future refinements—such as including inclination and distance uncertainties and tightening the mass-to-light priors—could clarify whether any residual preference for cored models persists.

5 Conclusions

Using a Bayesian model comparison across 175 SPARC galaxies, we find that cuspy NFW-like halos remain statistically preferred, with 85% of galaxies favoring either NFW or gNFW. The Bose–Einstein Condensate (BEC) and BEC-rotating profiles, while physically motivated and capable of producing cored profiles, achieve higher evidence in only $\sim 15\%$ of galaxies. This indicates that the cored inner density structures of galaxies can be captured phenomenologically through variable-slope NFW profiles without invoking condensate physics. Note that there are certain limitations to this research, such as the limited convergence diagnostics for all 175 galaxies (only 39 full posteriors analyzed).

These results support the robustness of the Λ CDM paradigm on galactic scales while leaving open the

possibility that condensate or wave-like dark matter may play a role in low-mass, diffuse systems. Future work incorporating inclination, distance priors, baryonic feedback, and non-circular motion uncertainties will clarify the residual parameter space where BEC-type models could be competitive.

References

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